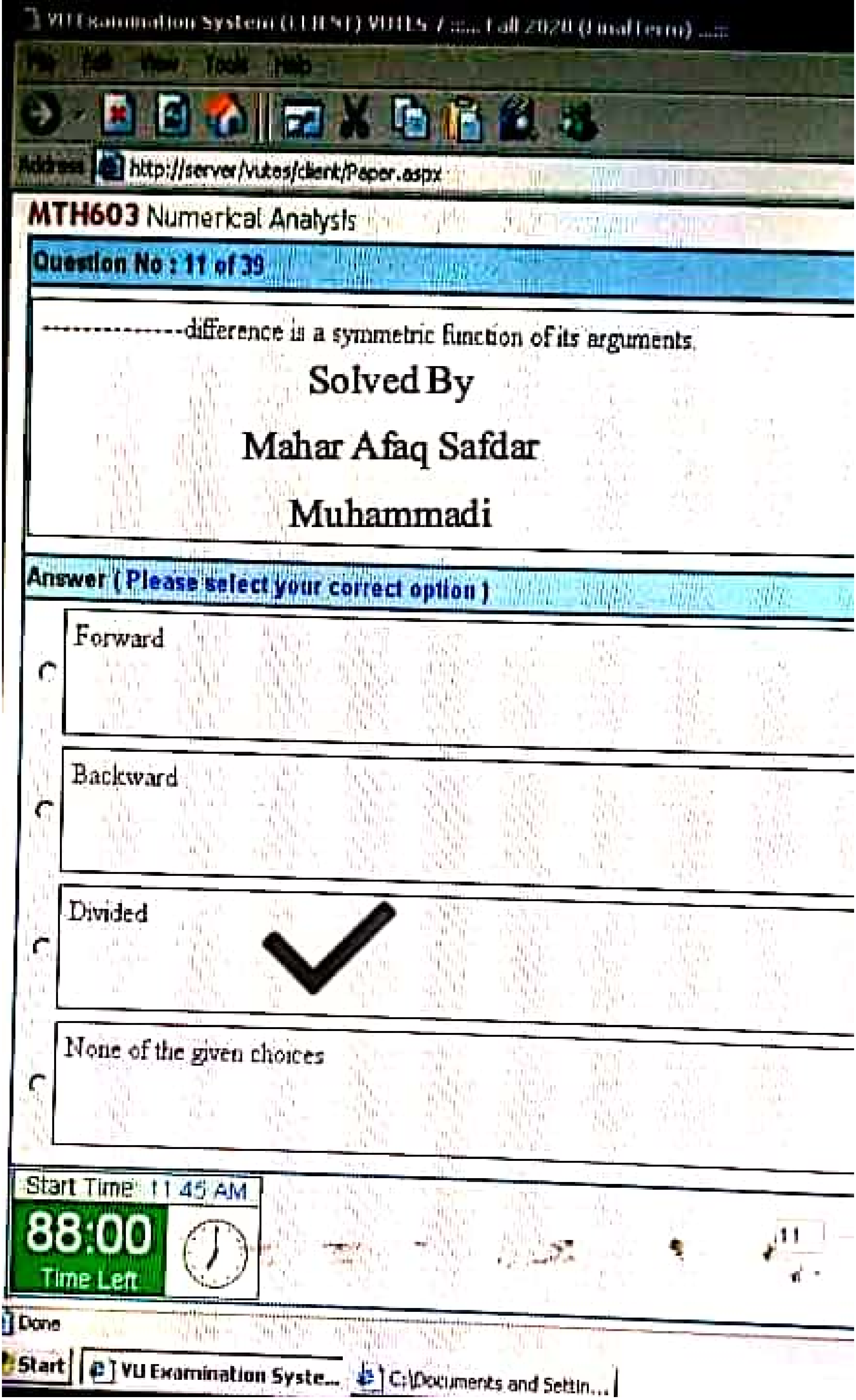
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MTH603 Numerical Analysis

Question No : 11 of 39

--------------difference is a symmetric function of its arguments.

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Muhammadi

Answer (**Please select your correct option**)

○ Forward

○ Backward

○ Divided ✓

○ None of the given choices

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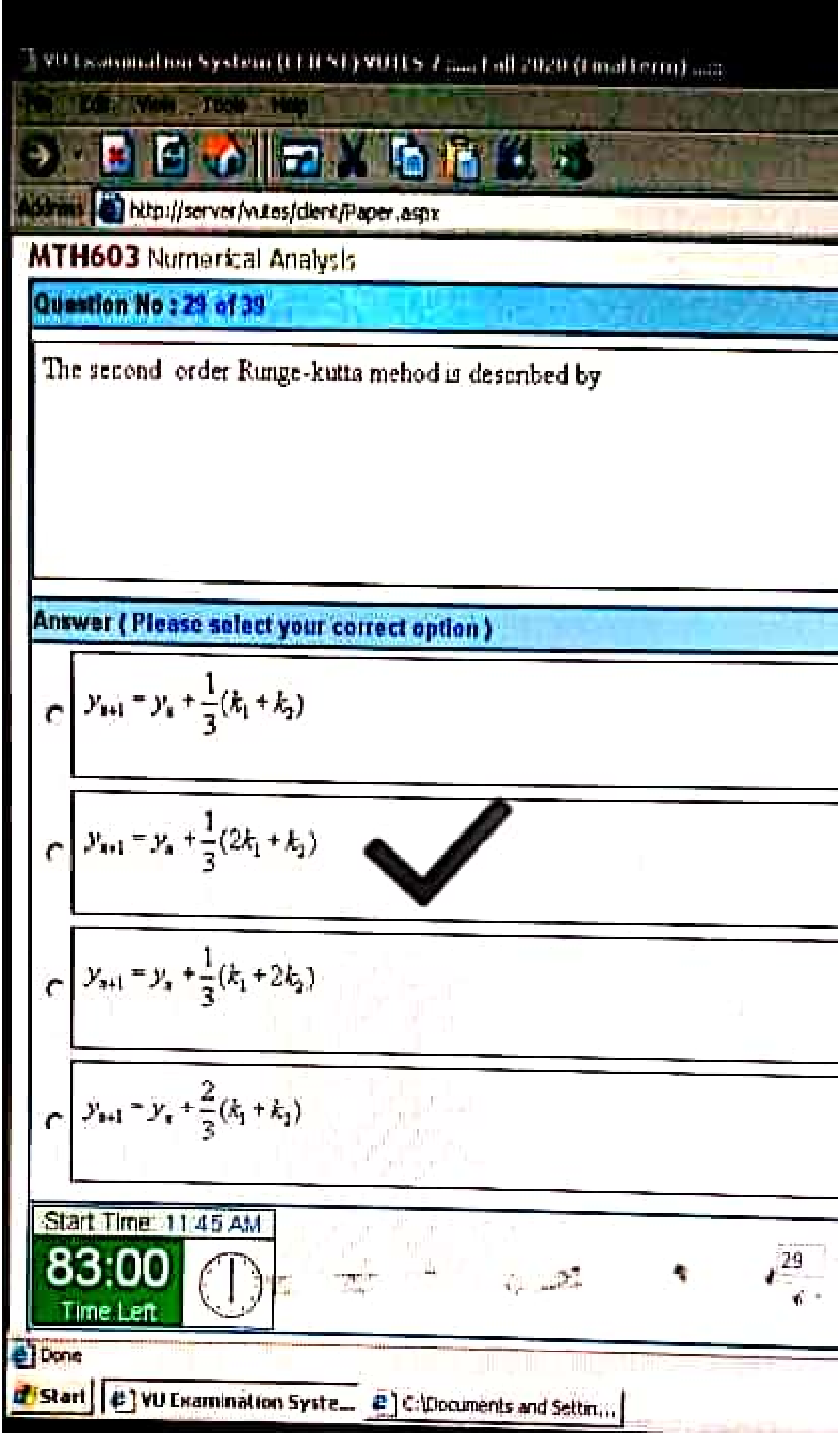

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MTH603 Numerical Analysis

Question No : 29 of 39

The second order Runge-kutta mehod is described by

Answer (Please select your correct option)

○ $y_{n+1} = y_n + \frac{1}{3}(k_1 + k_2)$

○ [✓ $y_{n+1} = y_n + \frac{1}{3}(2k_1 + k_2)$]

○ $y_{n+1} = y_n + \frac{1}{3}(k_1 + 2k_2)$

○ $y_{n+1} = y_n + \frac{2}{3}(k_1 + k_2)$

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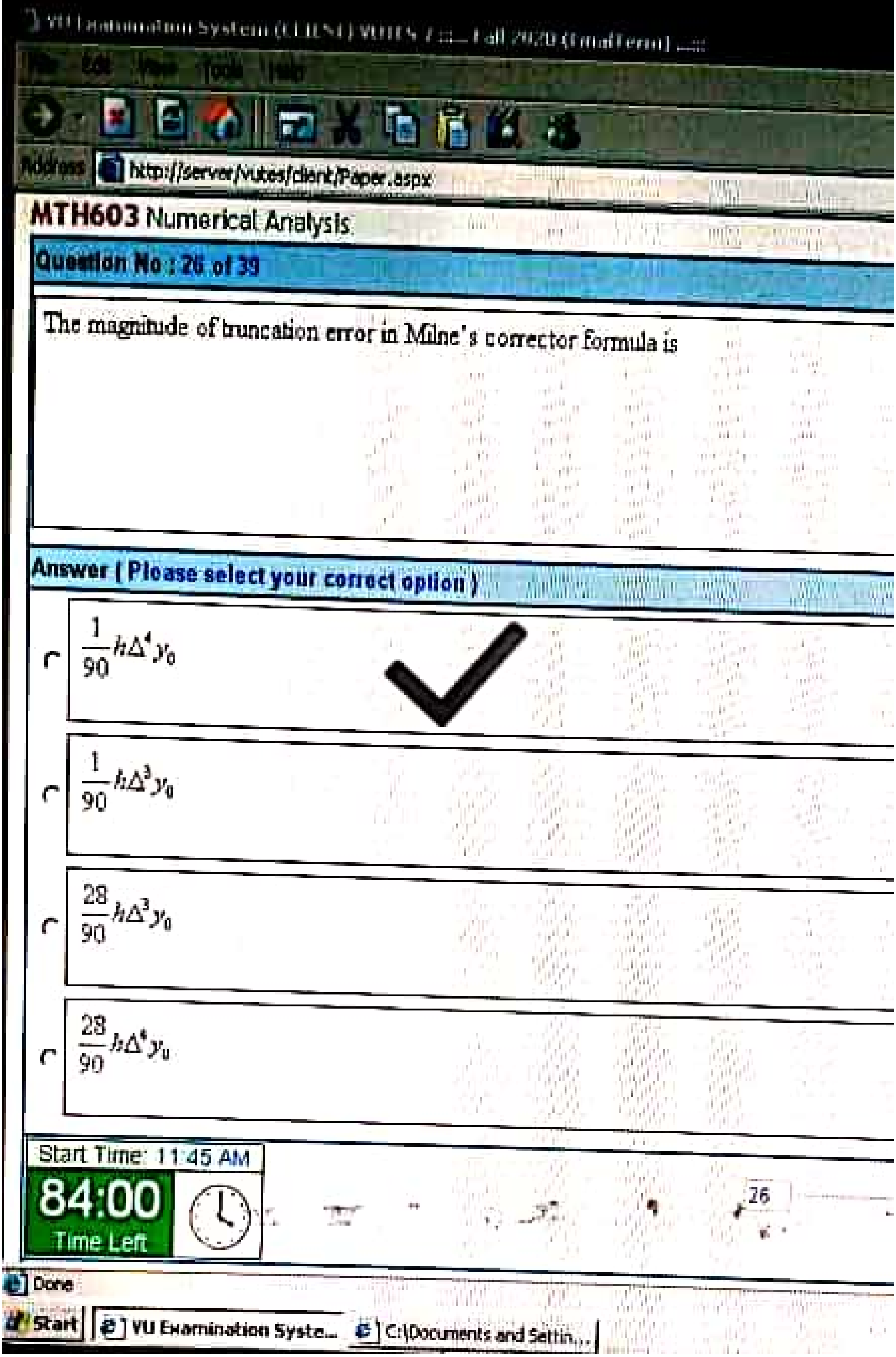
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MTH603 Numerical Analysis

Question No : 26 of 39

The magnitude of truncation error in Milne's corrector formula is

Answer (Please select your correct option)

○ $\frac{1}{90}h\Delta^4 y_0$ ✓

○ $\frac{1}{90}h\Delta^3 y_0$

○ $\frac{28}{90}h\Delta^3 y_0$

○ $\frac{28}{90}h\Delta^4 y_0$

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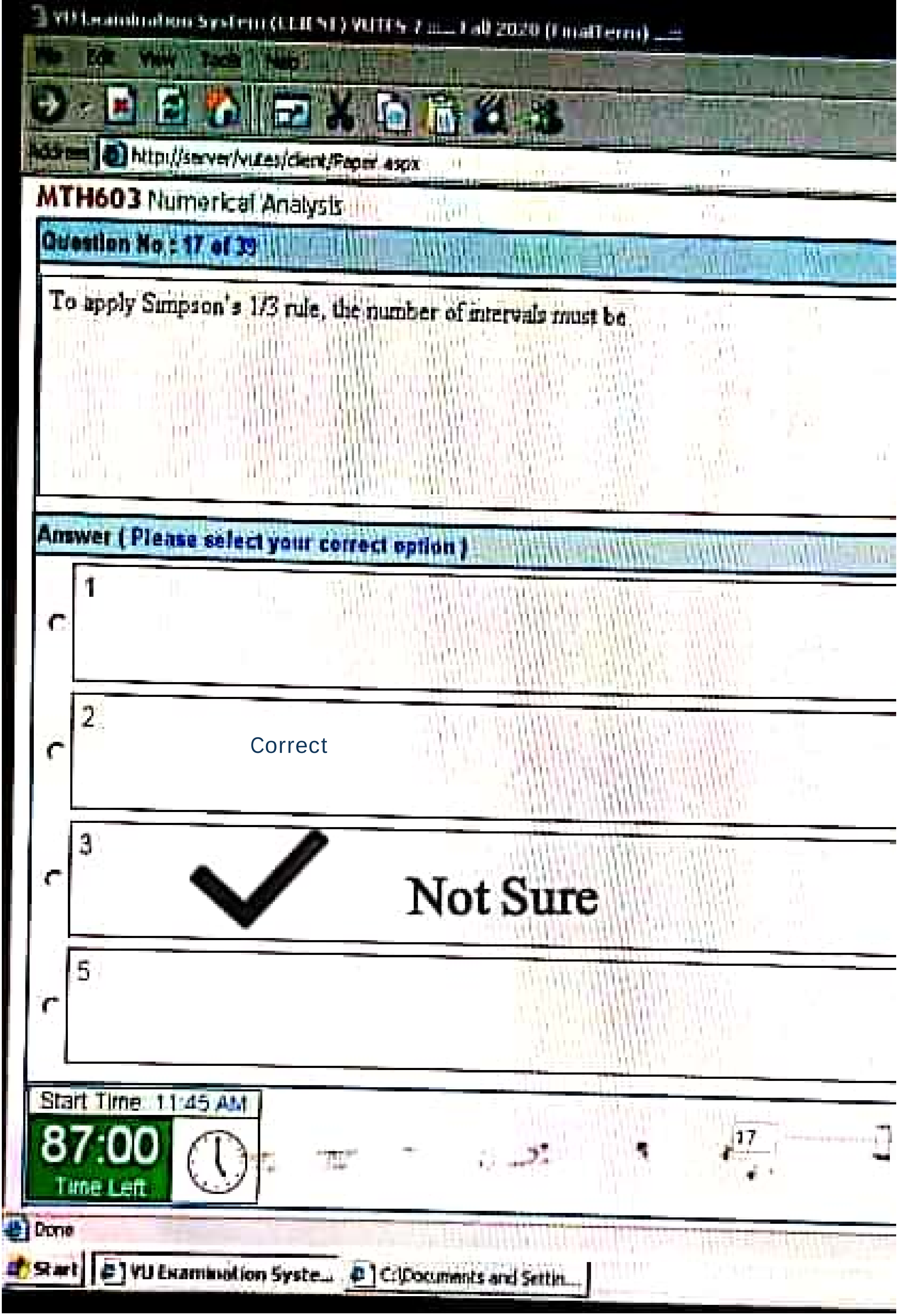
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MTH603 Numerical Analysis

Question No : 17 of 39

To apply Simpson's 1/3 rule, the number of intervals must be

Answer (Please select your correct option)

○ 1

○ 2 Correct

○ 3 ✓ **Not Sure**

○ 5

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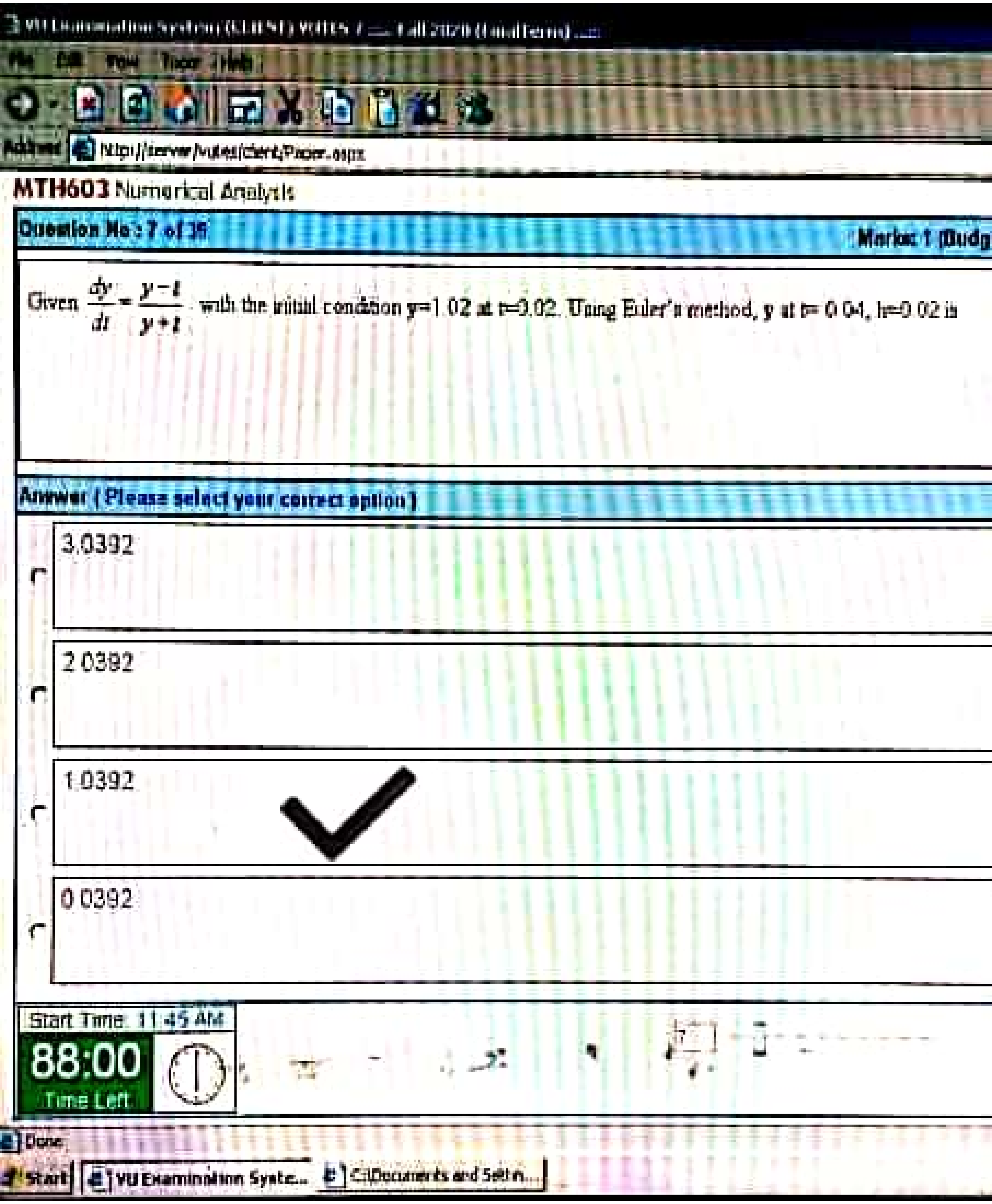
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MTH603 Numerical Analysis

Question No : 7 of 35

Marks: 1 (Budg

Given $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial condition y=1.02 at t=0.02. Using Euler's method, y at t= 0.04, h=0.02 is

Answer (**Please select your correct option**)

○ 3.0392

○ 2.0392

○ [✓ 1.0392]

○ 0.0392

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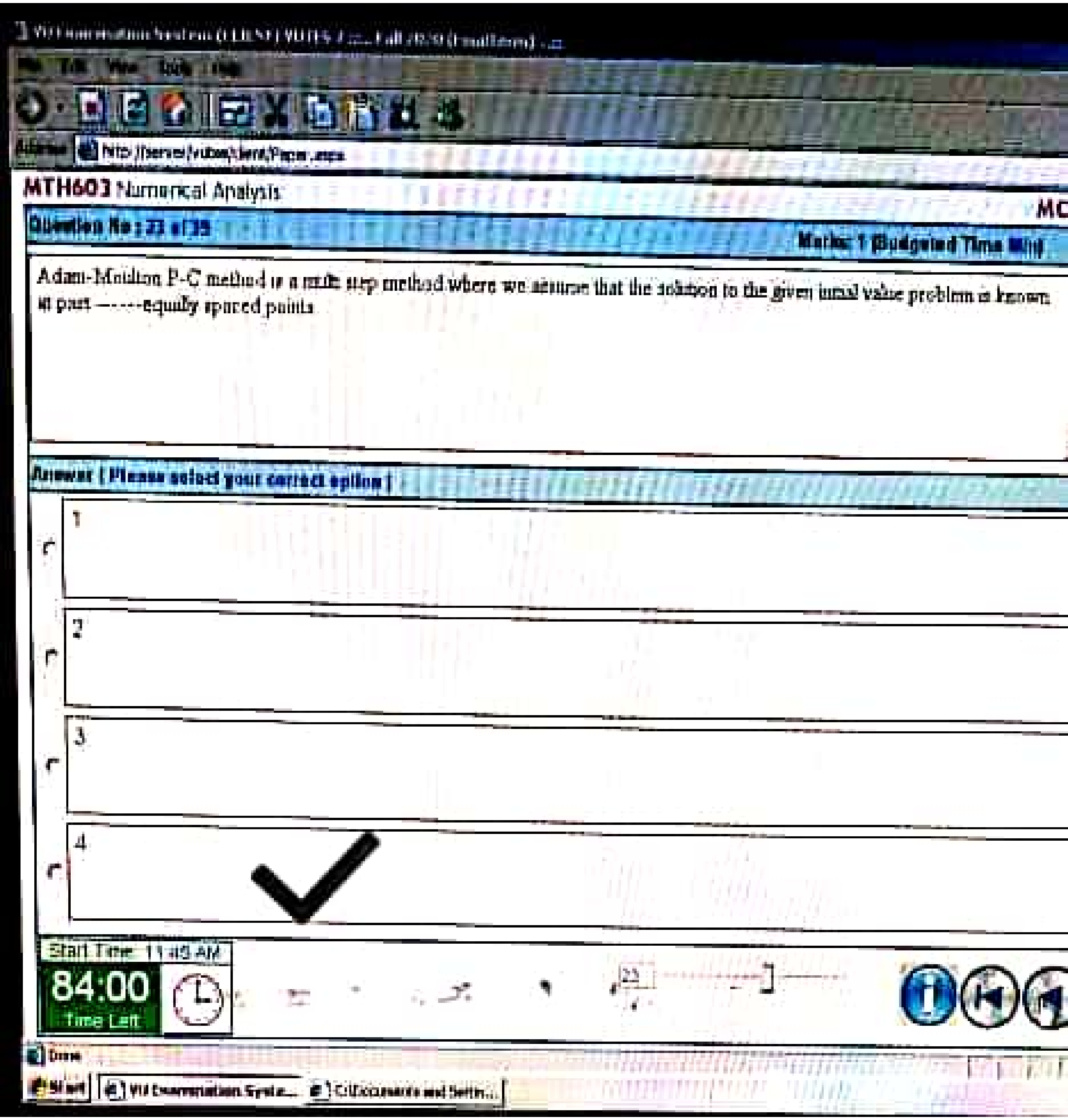
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MTH603 Numerical Analysis

Question No : 23 of 35

Marks: 1 (Budgeted Time

Adam-Moulton P-C method is a multi step method where we assume that the solution to the given initial value problem is known at past ------equally spaced points

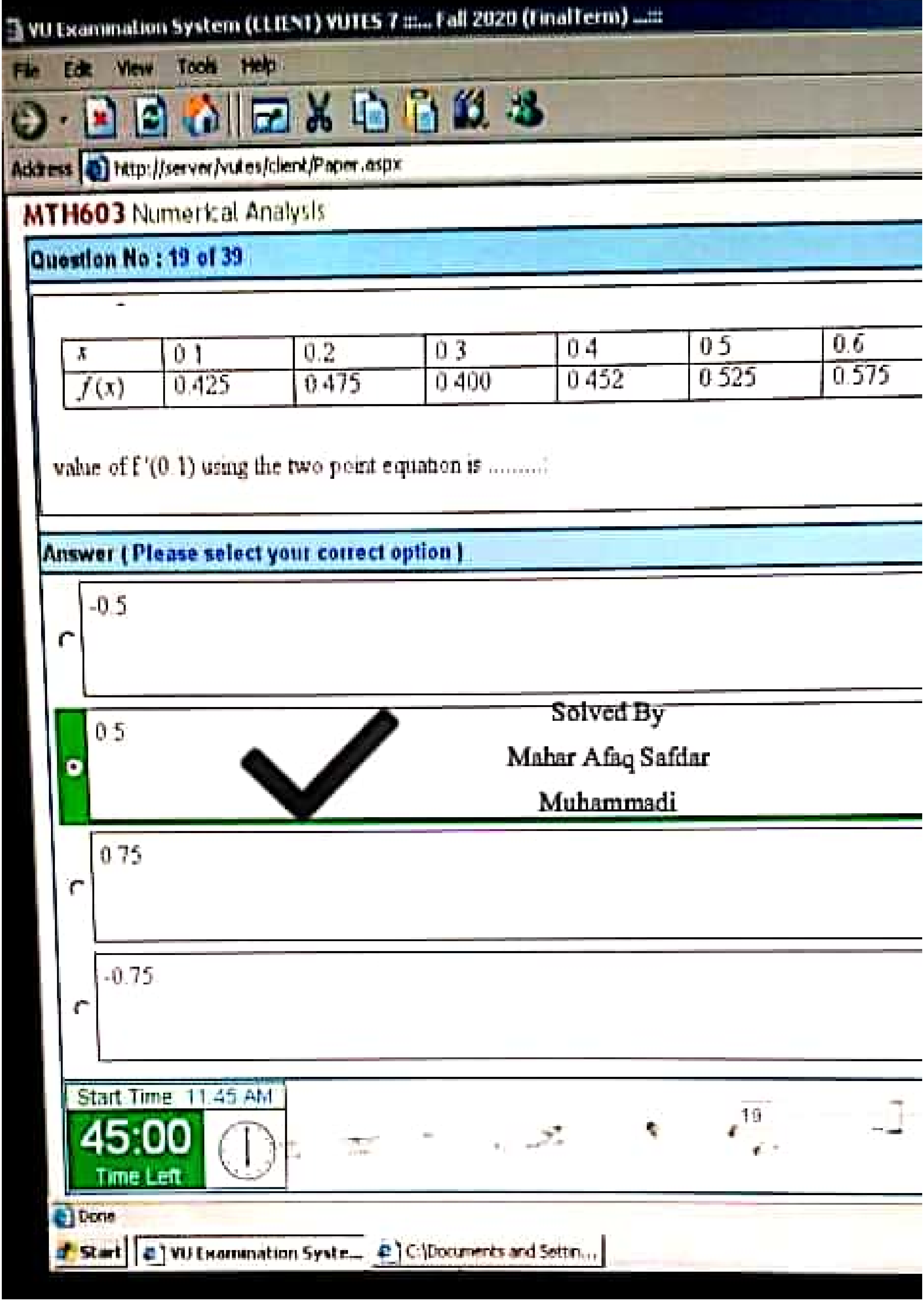
Answer (Please select your correct option)

○ 1

○ 2

○ 3

○ 4 ✓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MTH603 Numerical Analysis

Question No : 19 of 39

x	0.1	0.2	0.3	0.4	0.5	0.6
$f(x)$	0.425	0.475	0.400	0.452	0.525	0.575

value of f '(0.1) using the two point equation is ………

Answer (Please select your correct option)

○ -0.5

● 0.5 ✓

○ 0.75

○ -0.75

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Muhammadi

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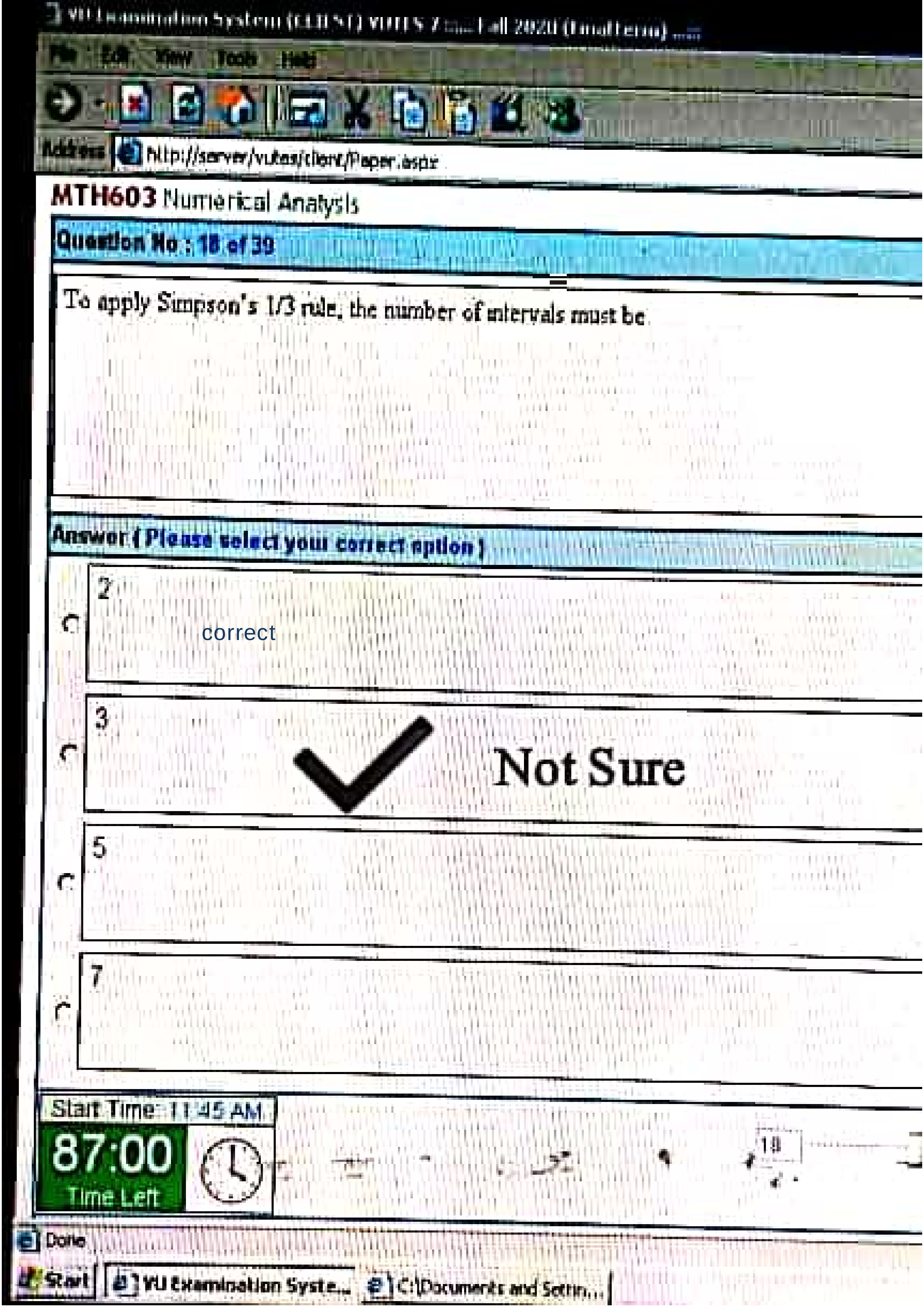
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MTH603 Numerical Analysis

Question No : 18 of 39

To apply Simpson's 1/3 rule, the number of intervals must be

Answer (Please select your correct option)

○ 2

correct

○ 3

✓ Not Sure

○ 5

○ 7

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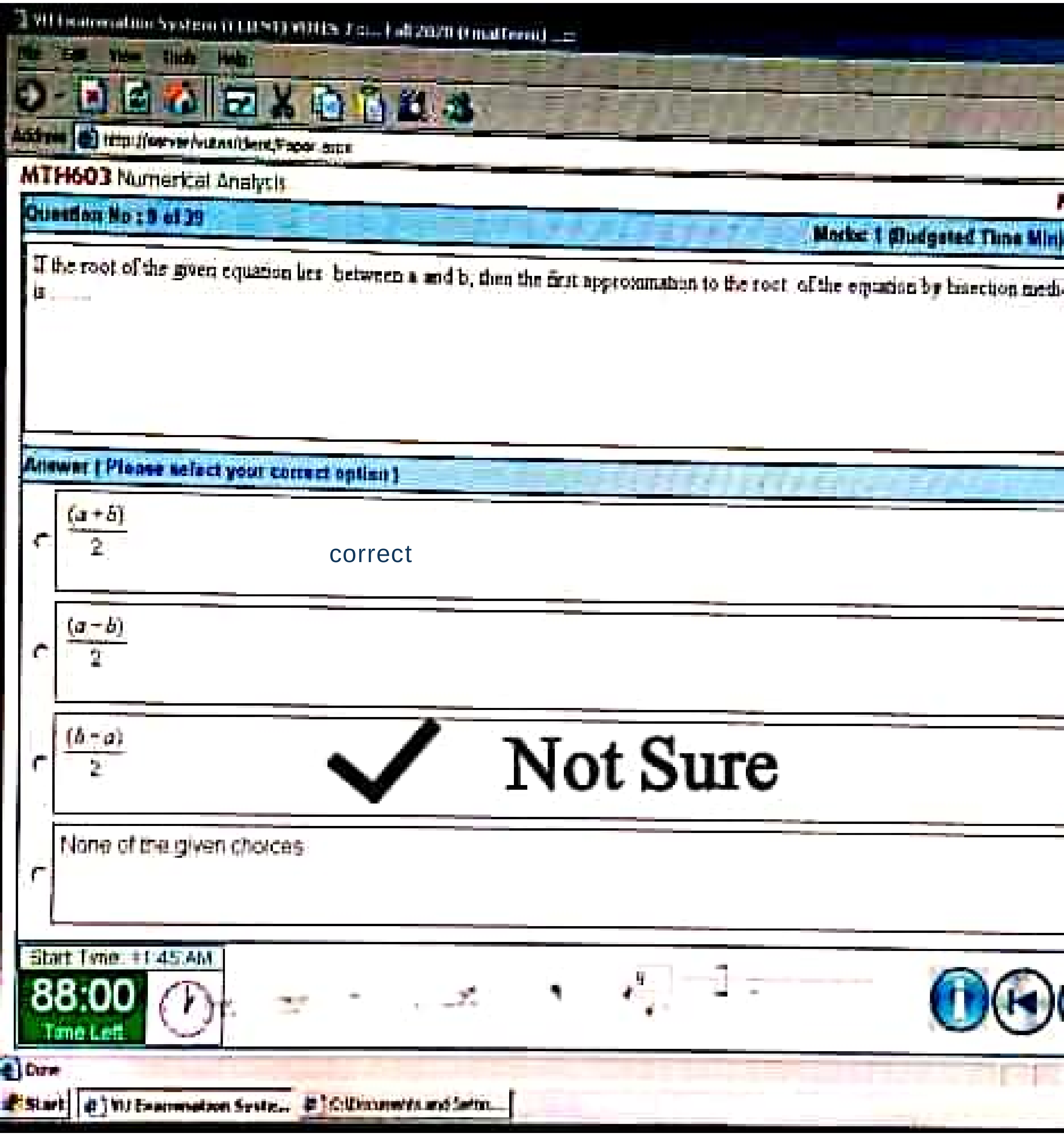

MTH603 Numerical Analysis

Question No : 9 of 39

Marks: 1 (Budgeted Time Min)

If the root of the given equation lies between a and b, then the first approximation to the root of the equation by bisection medi
is ......

Answer (Please select your correct option)

○ $\frac{(a+b)}{2}$

correct

○ $\frac{(a-b)}{2}$

○ $\frac{(b-a)}{2}$

✓ **Not Sure**

○ None of the given choices

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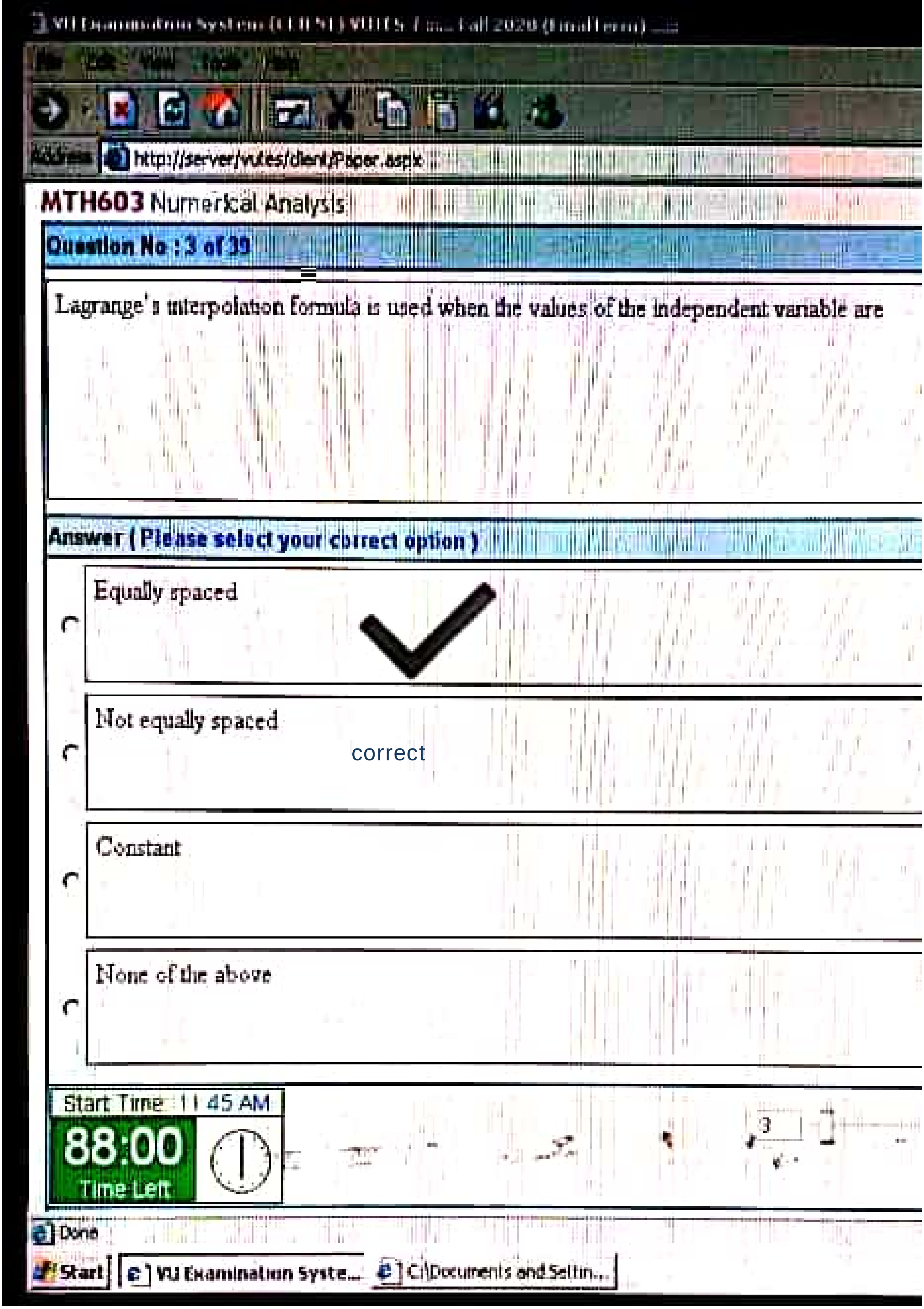
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MTH603 Numerical Analysis

Question No : 3 of 39

Lagrange's interpolation formula is used when the values of the independent variable are

Answer (Please select your correct option)

○ Equally spaced ✓

○ Not equally spaced

correct

○ Constant

○ None of the above

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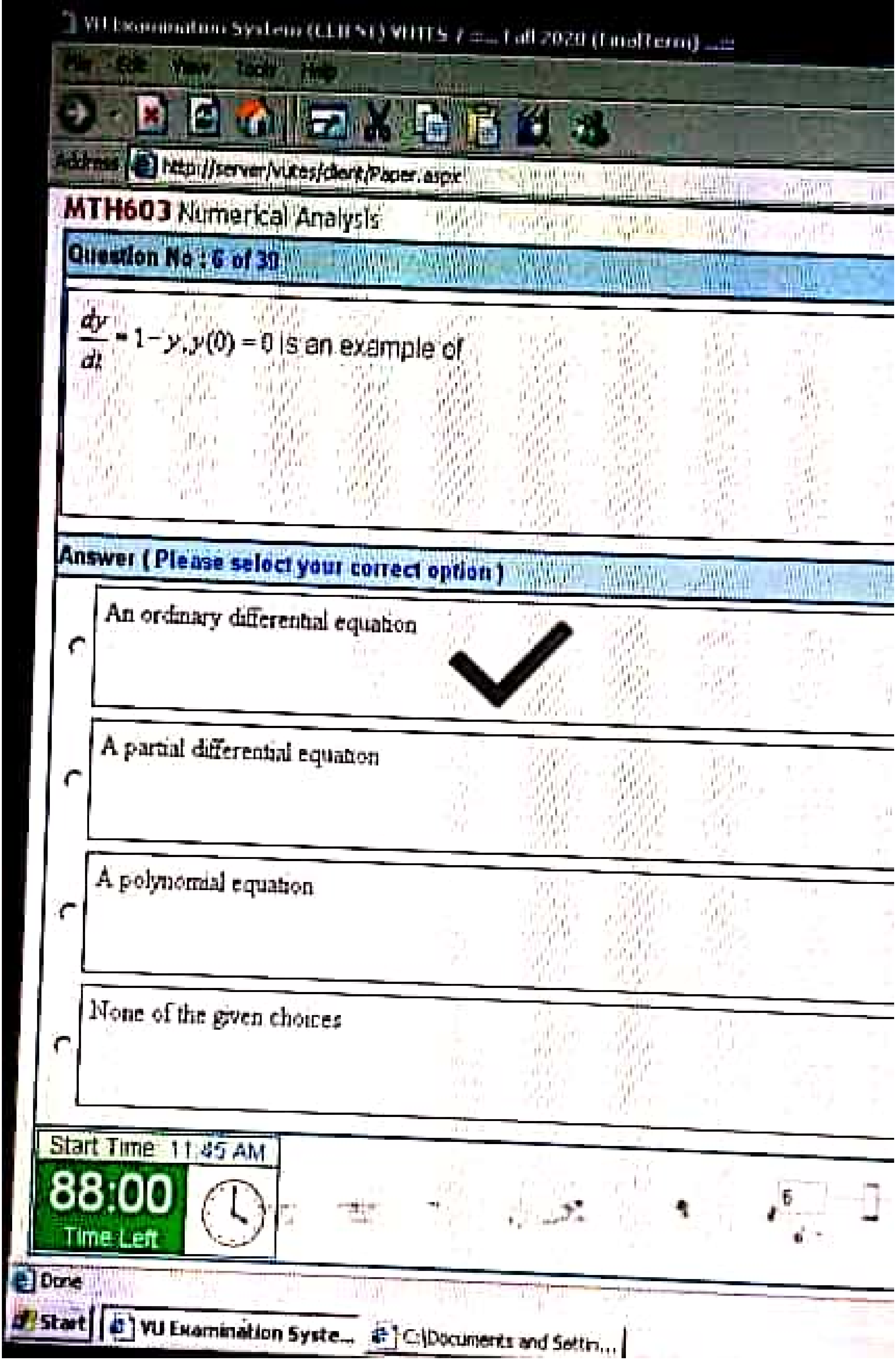
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MTH603 Numerical Analysis

Question No : 6 of 30

$\frac{dy}{dt} = 1 - y, y(0) = 0$ is an example of

Answer (Please select your correct option)

○ An ordinary differential equation ✓

○ A partial differential equation

○ A polynomial equation

○ None of the given choices

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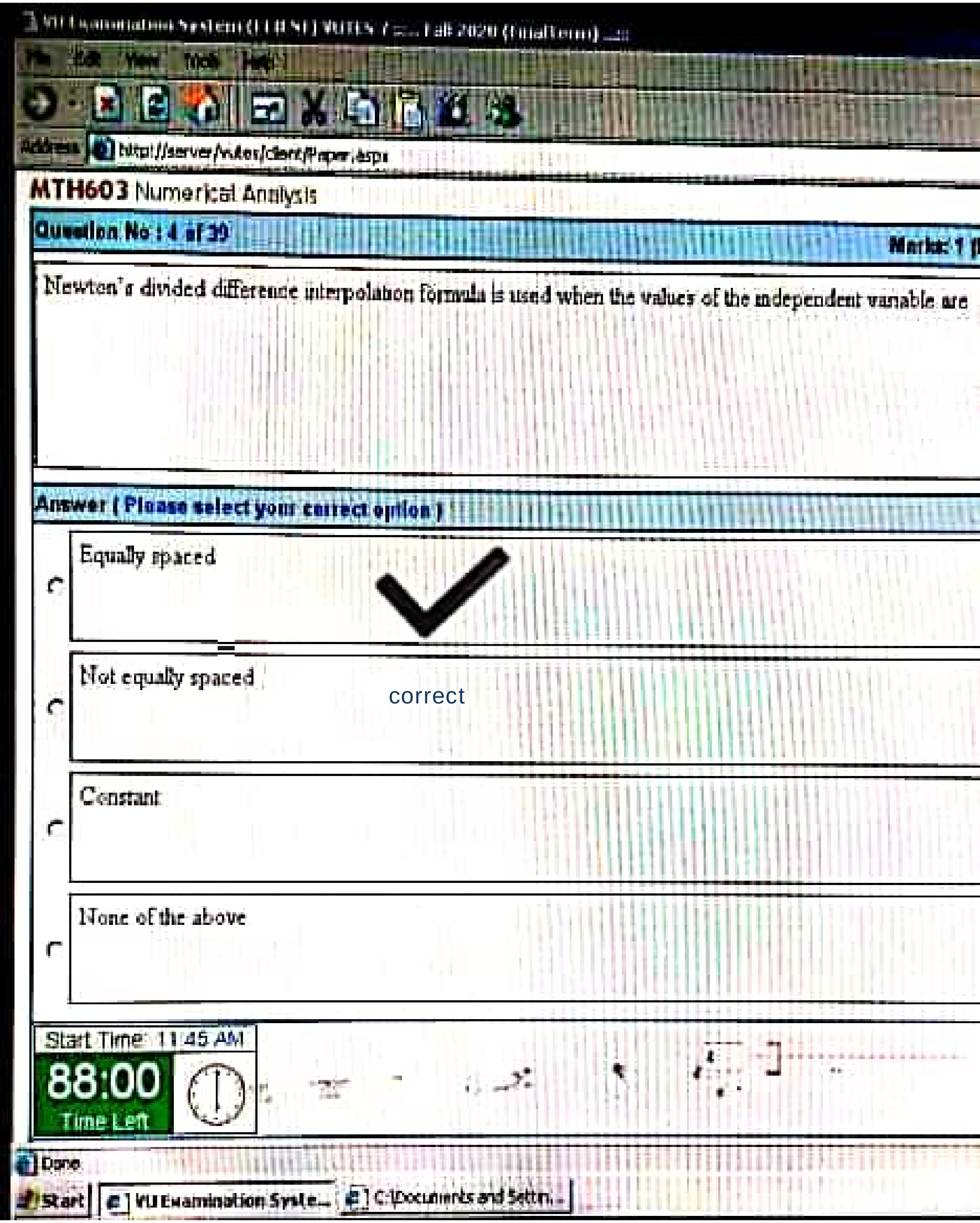
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MTH603 Numerical Analysis

Question No : 4 of 30 **Marks: 1**

Newton's divided difference interpolation formula is used when the values of the independent variable are

Answer (Please select your correct option)

○ Equally spaced ✓

○ Not equally spaced

correct

○ Constant

○ None of the above

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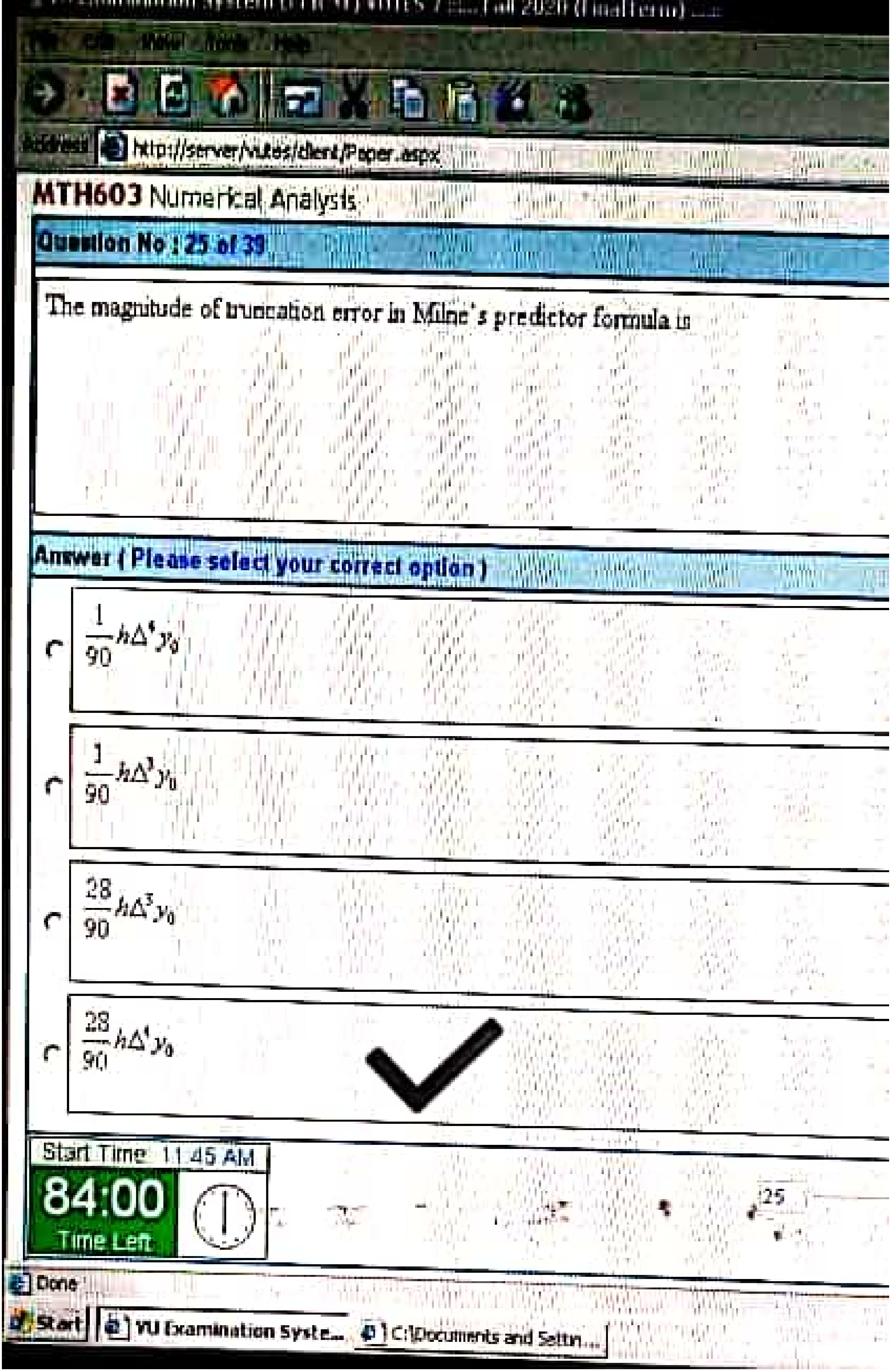
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MTH603 Numerical Analysis

Question No : 25 of 39

The magnitude of truncation error in Milne's predictor formula is

Answer (Please select your correct option)

○ $\frac{1}{90}h\Delta^4 y_0$

○ $\frac{1}{90}h\Delta^3 y_0$

○ $\frac{28}{90}h\Delta^3 y_0$

○ $\frac{28}{90}h\Delta^4 y_0$ ✓

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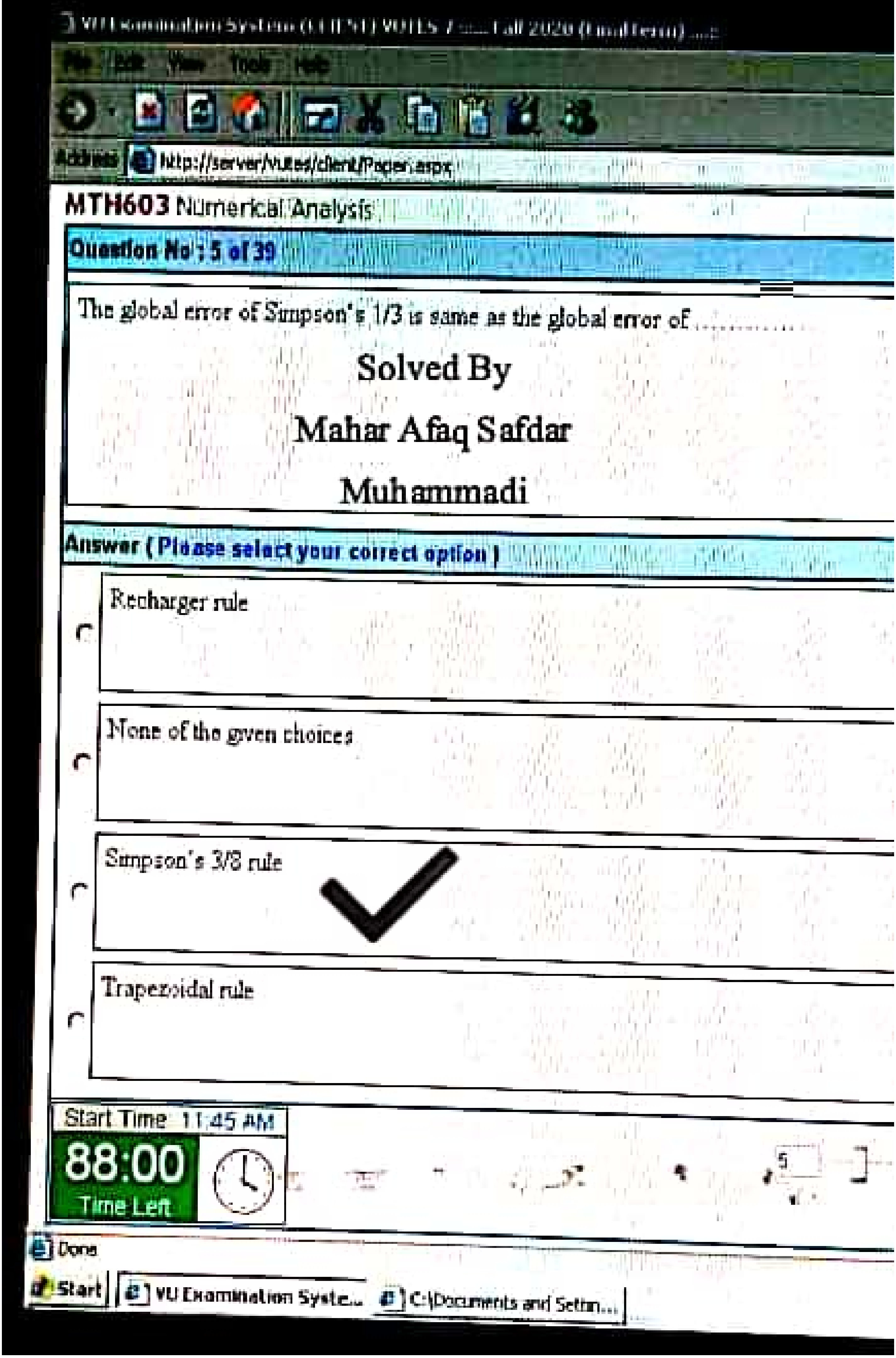
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MTH603 Numerical Analysis

Question No : 5 of 39

The global error of Simpson's 1/3 is same as the global error of ....

Solved By

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Muhammadi

Answer (Please select your correct option)

○ Recharger rule

○ None of the given choices

○ Simpson's 3/8 rule ✓

○ Trapezoidal rule

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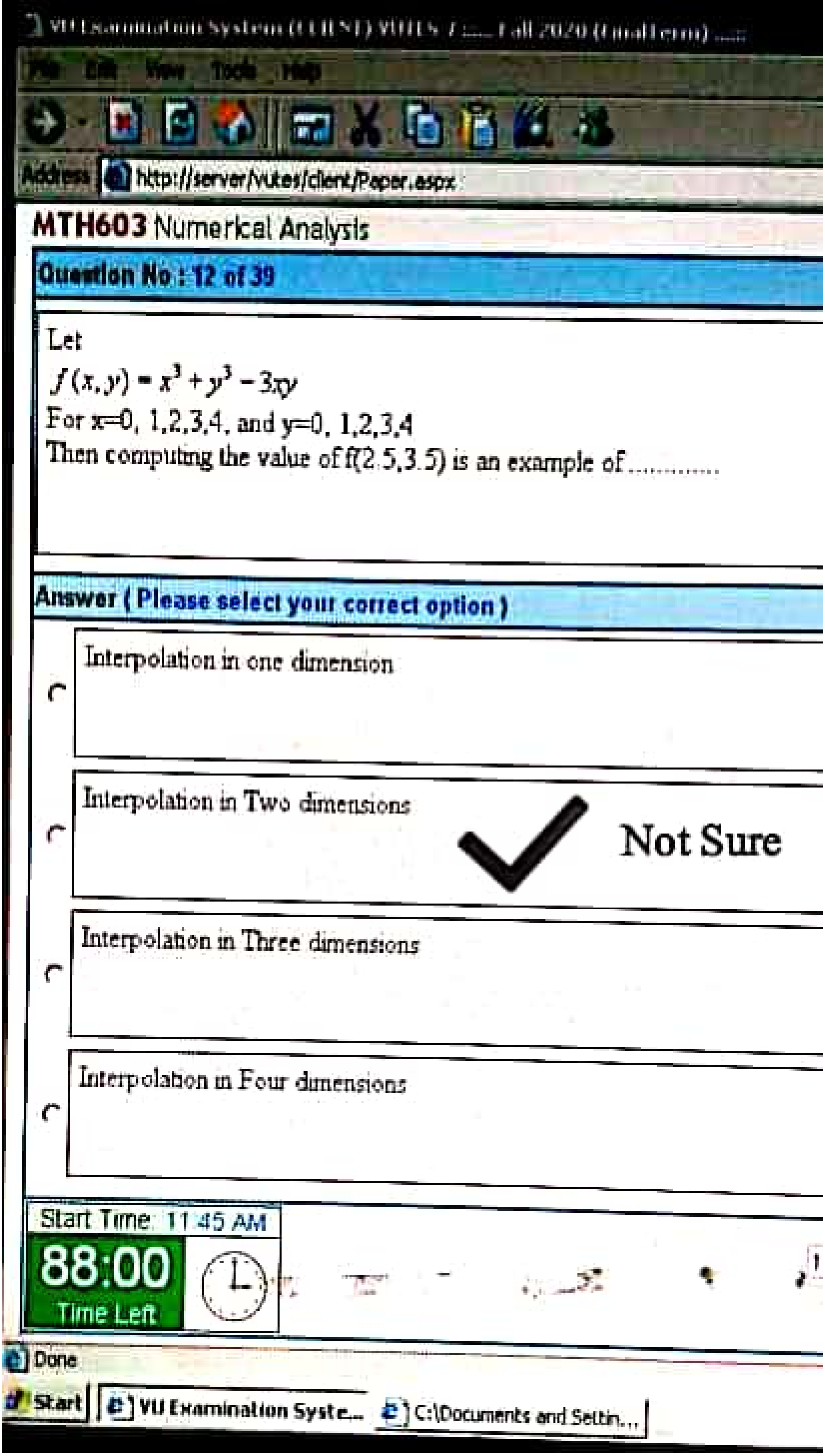

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MTH603 Numerical Analysis

Question No : 12 of 39

Let

$f(x,y) = x^3 + y^3 - 3xy$

For x=0, 1,2,3,4, and y=0, 1,2,3,4

Then computing the value of f(2.5,3.5) is an example of ....

Answer (Please select your correct option)

○ Interpolation in one dimension

○ Interpolation in Two dimensions ✓ Not Sure

○ Interpolation in Three dimensions

○ Interpolation in Four dimensions

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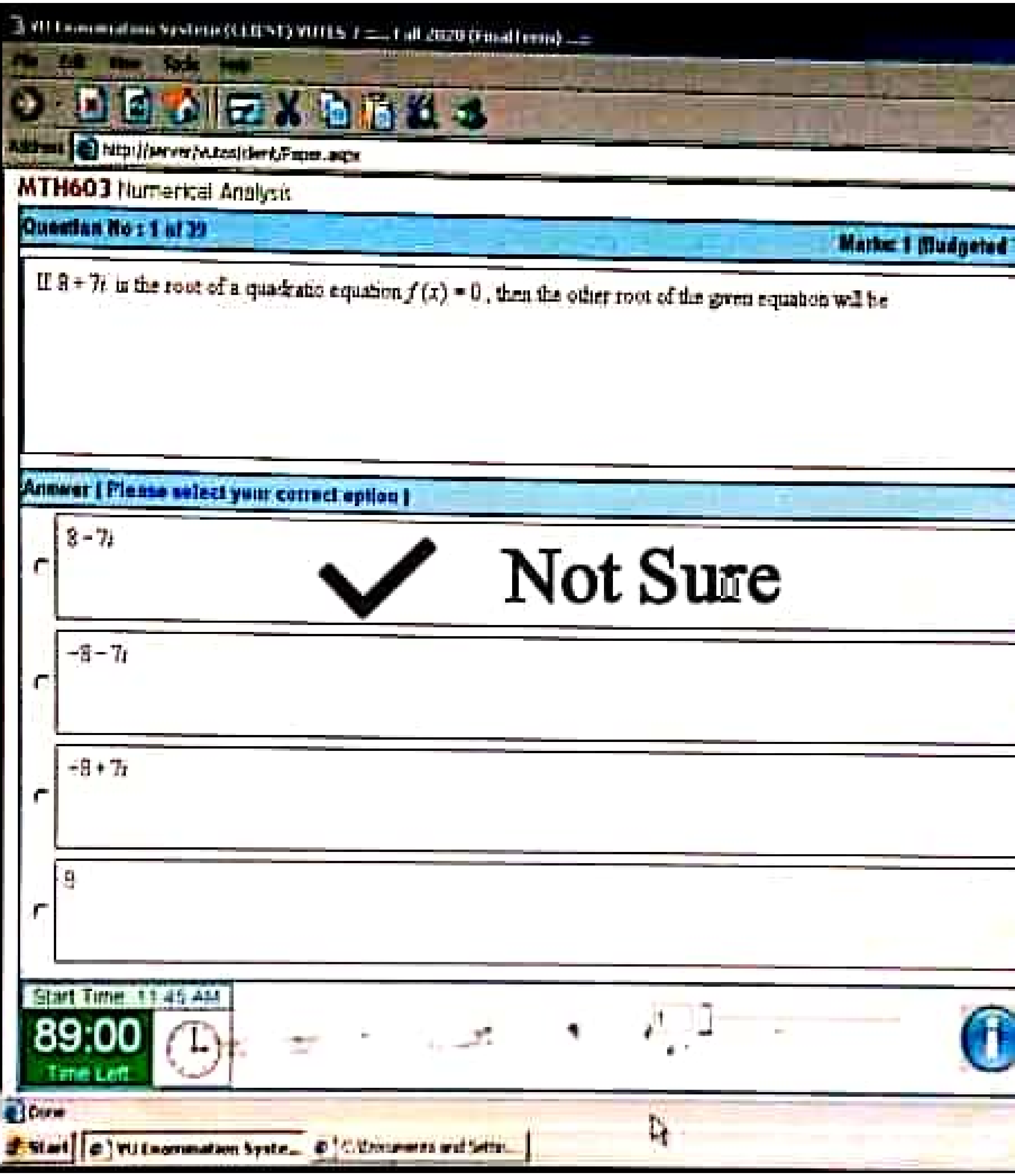

MTH603 Numerical Analysis

Question No : 1 of 39

Marks: 1 (Budgeted

If $8 + 7i$ is the root of a quadratic equation $f(x) = 0$, then the other root of the given equation will be

Answer (Please select your correct option)

○ $8 - 7i$ ✓ Not Sure

○ $-8 - 7i$

○ $-8 + 7i$

○ 9

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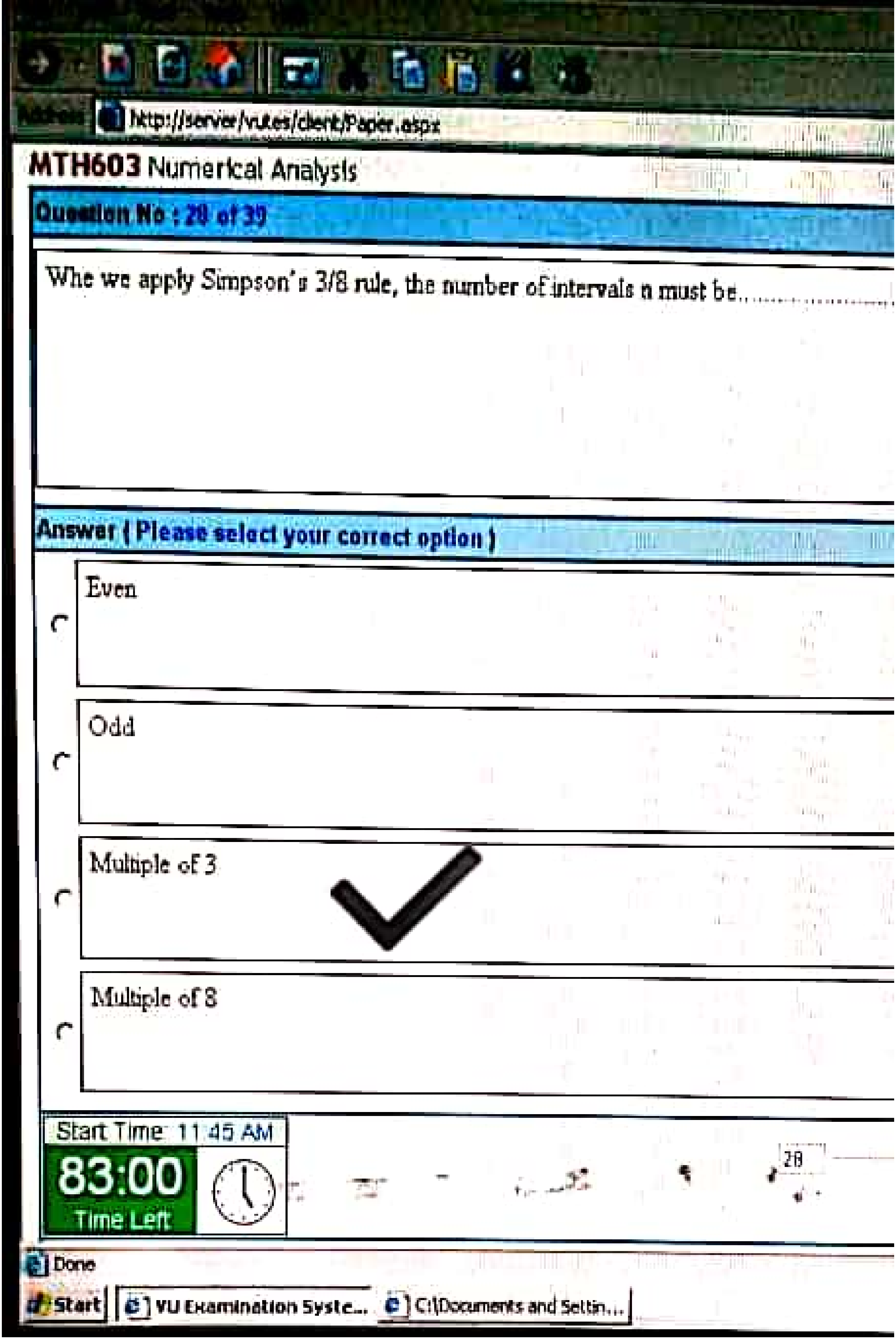

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MTH603 Numerical Analysis

Question No : 28 of 39

Whe we apply Simpson's 3/8 rule, the number of intervals n must be.......................

Answer (Please select your correct option)

○ Even

○ Odd

○ [✓ Multiple of 3]

○ Multiple of 8

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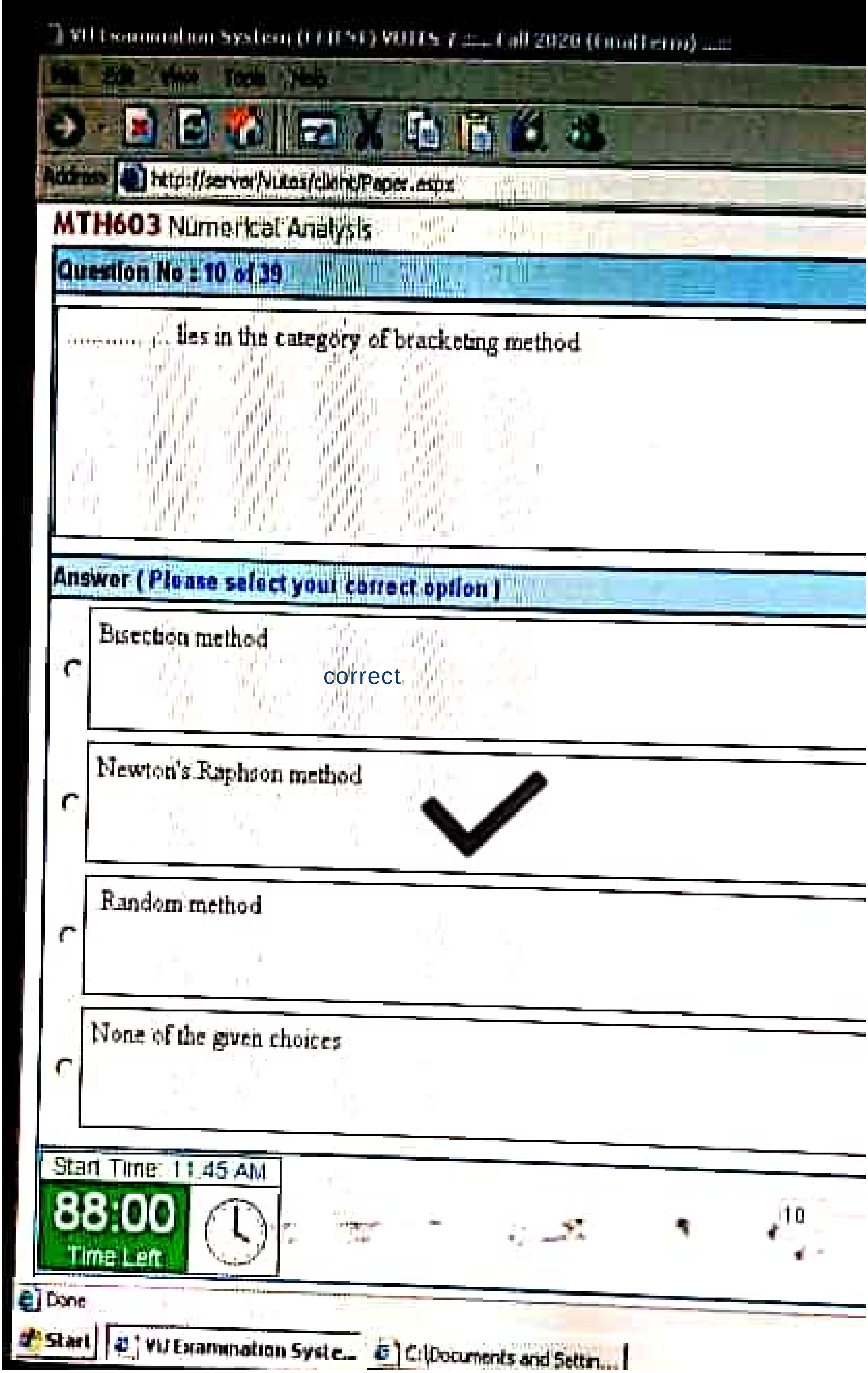
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MTH603 Numerical Analysis

Question No : 10 of 39

.... lies in the category of bracketing method

Answer (Please select your correct option)

○ Bisection method

correct

○ Newton's Raphson method ✓

○ Random method

○ None of the given choices

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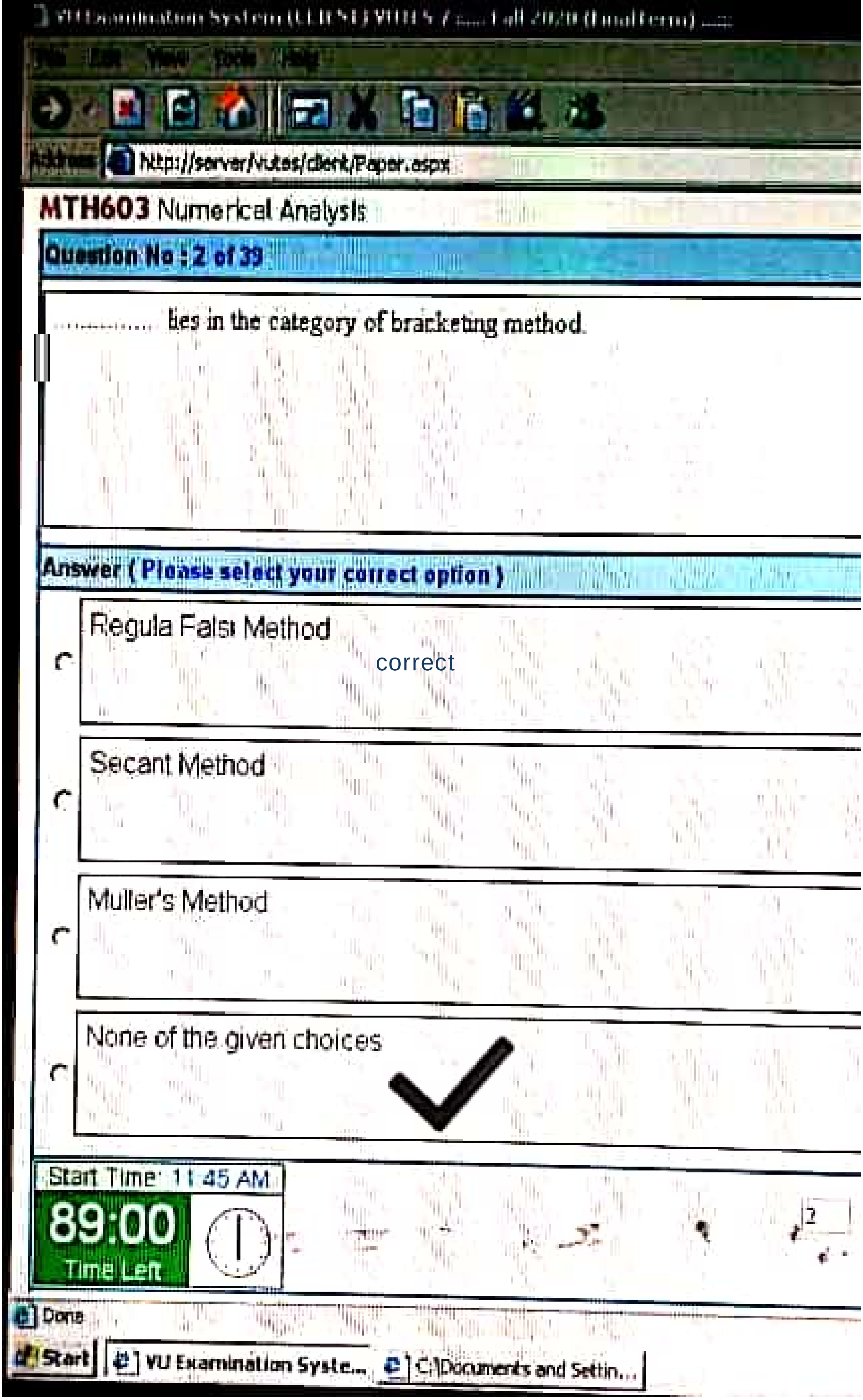

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MTH603 Numerical Analysis

Question No : 2 of 39

………….. lies in the category of bracketing method.

Answer (**Please select your correct option**)

○ Regula Falsi Method

correct

○ Secant Method

○ Muller's Method

○ None of the given choices ✓

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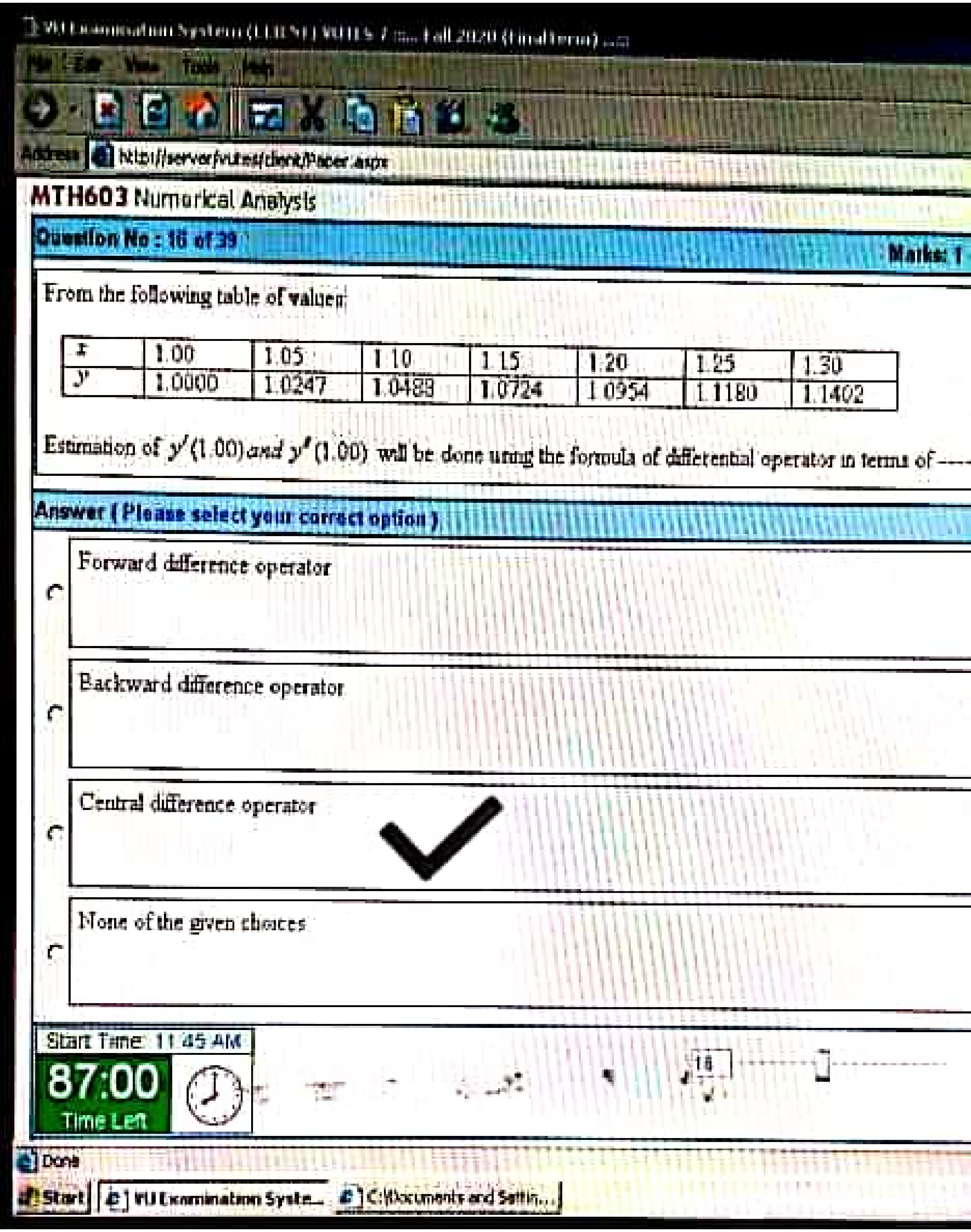

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MTH603 Numerical Analysis

Question No : 16 of 39 **Marks: 1**

From the following table of values

x	1.00	1.05	1.10	1.15	1.20	1.25	1.30
y	1.0000	1.0247	1.0488	1.0724	1.0954	1.1180	1.1402

Estimation of $y'(1.00)$ *and* $y''(1.00)$ will be done using the formula of differential operator in terms of ----

Answer (Please select your correct option)

○ Forward difference operator

○ Backward difference operator

○ Central difference operator ✓

○ None of the given choices

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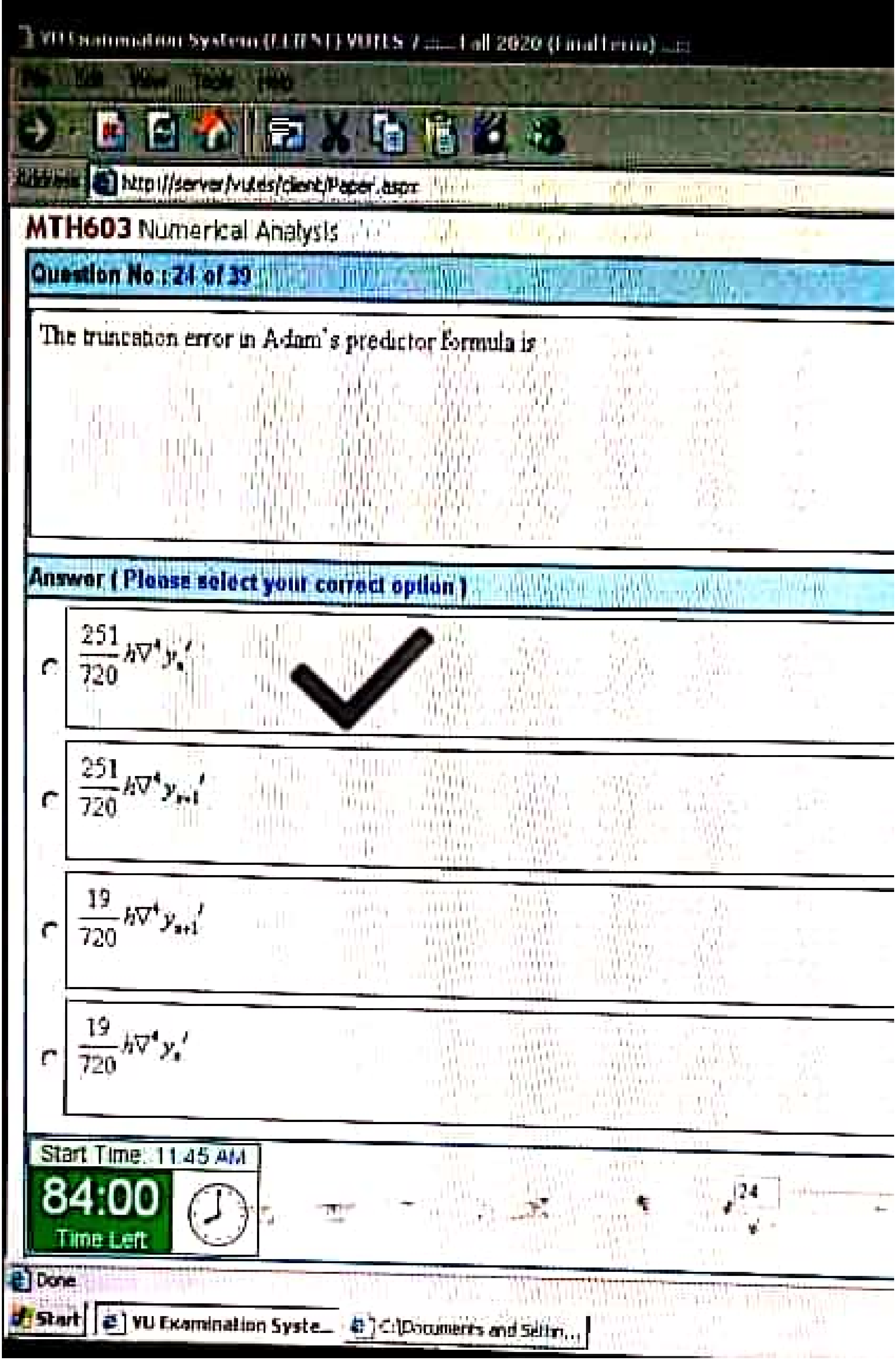
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MTH603 Numerical Analysis

Question No : 24 of 39

The truncation error in Adam's predictor formula is

Answer (Please select your correct option)

○ [✓ $\frac{251}{720} h\nabla^4 y_n'$]

○ $\frac{251}{720} h\nabla^4 y_{n+1}'$

○ $\frac{19}{720} h\nabla^4 y_{n+1}'$

○ $\frac{19}{720} h\nabla^4 y_n'$

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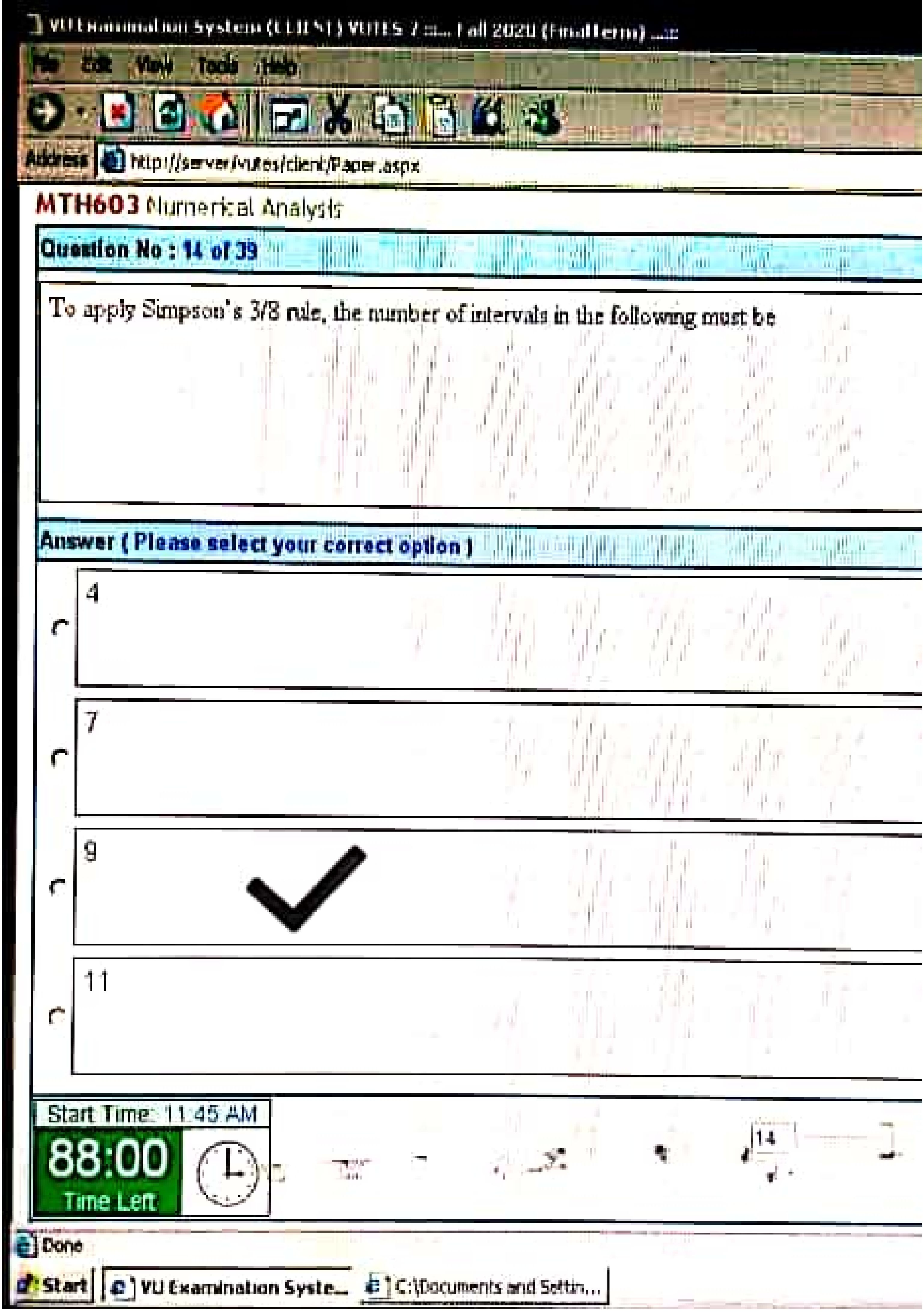
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MTH603 Numerical Analysis

Question No : 14 of 39

To apply Simpson's 3/8 rule, the number of intervals in the following must be

Answer (Please select your correct option)

○ 4

○ 7

○ 9 ✓

○ 11

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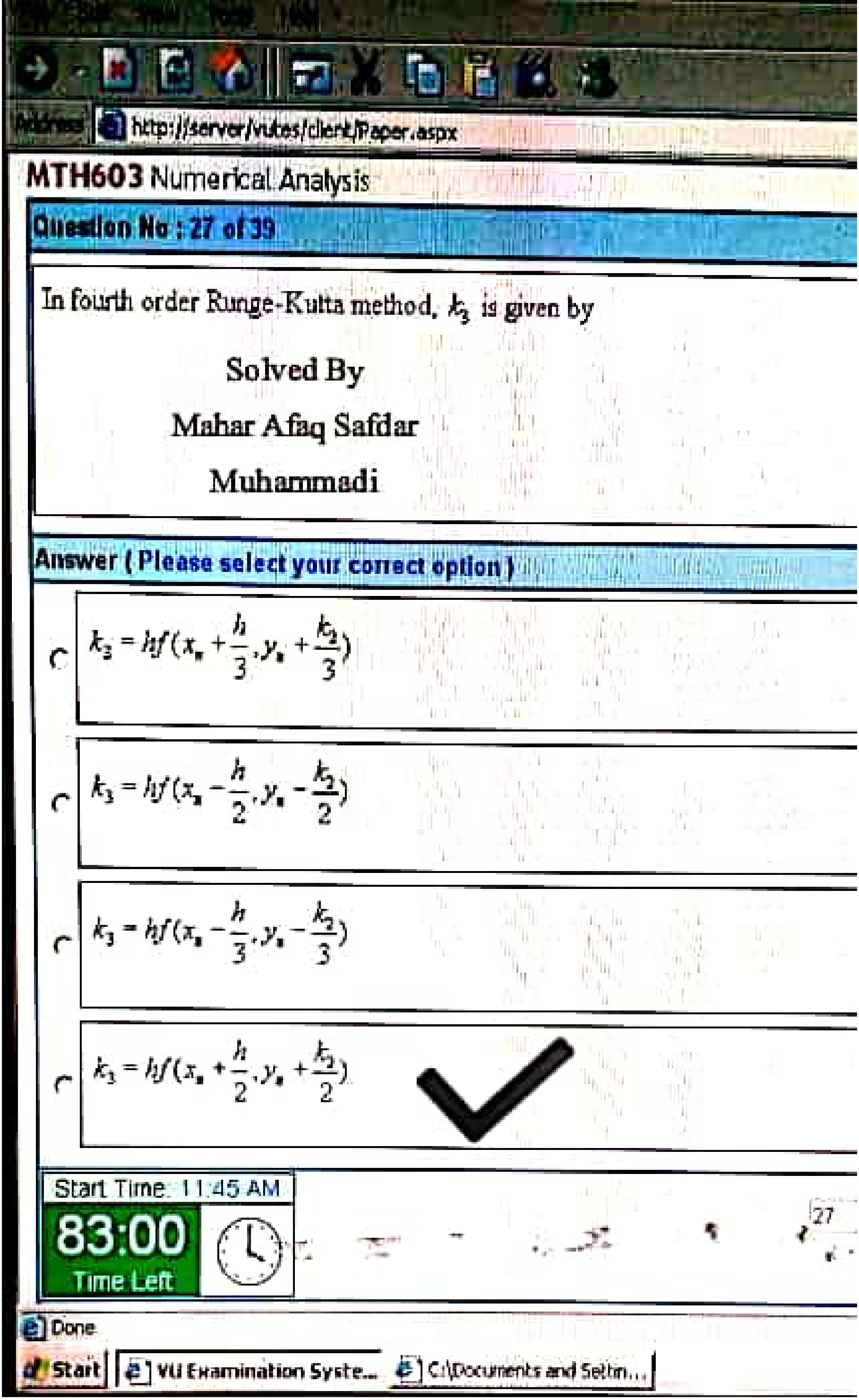

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MTH603 Numerical Analysis

Question No : 27 of 39

In fourth order Runge-Kutta method, k_3 is given by

Solved By

Mahar Afaq Safdar

Muhammadi

Answer (**Please select your correct option**)

○ $k_3 = hf(x_n + \frac{h}{3}, y_n + \frac{k_2}{3})$

○ $k_3 = hf(x_n - \frac{h}{2}, y_n - \frac{k_2}{2})$

○ $k_3 = hf(x_n - \frac{h}{3}, y_n - \frac{k_2}{3})$

○ $k_3 = hf(x_n + \frac{h}{2}, y_n + \frac{k_2}{2})$ ✓

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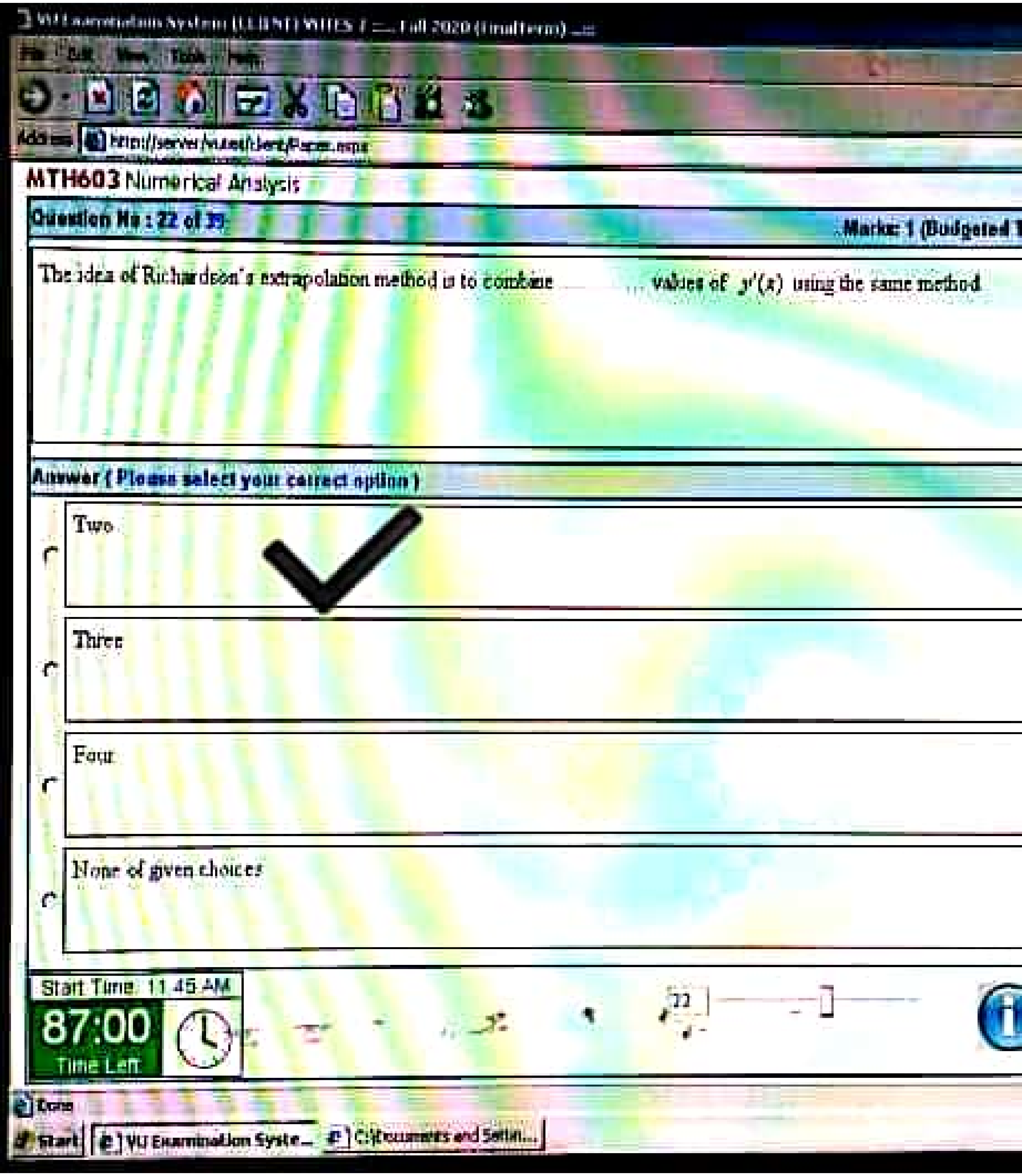
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MTH603 Numerical Analysis

Question No : 22 of 35 **Marks: 1 (Budgeted T**

The idea of Richardson's extrapolation method is to combine .... values of $y'(x)$ using the same method

Answer (Please select your correct option)

○ [✓ Two]

○ Three

○ Four

○ None of given choices

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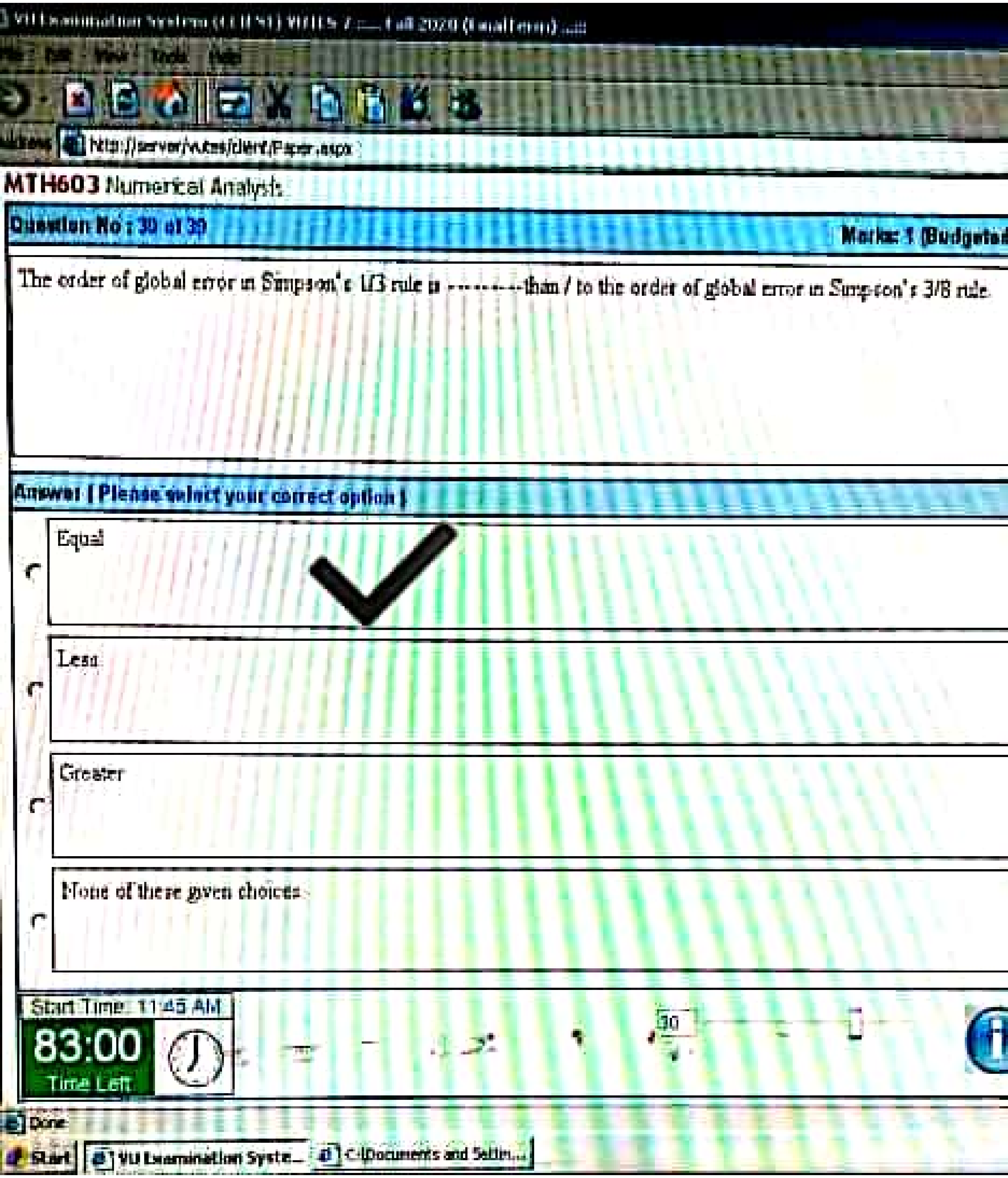
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MTH603 Numerical Analysis

Question No : 30 of 30 **Marks: 1 (Budgeted**

The order of global error in Simpson's 1/3 rule is ---------than / to the order of global error in Simpson's 3/8 rule.

Answer (Please select your correct option)

○ [✓ Equal]

○ Less

○ Greater

○ None of these given choices

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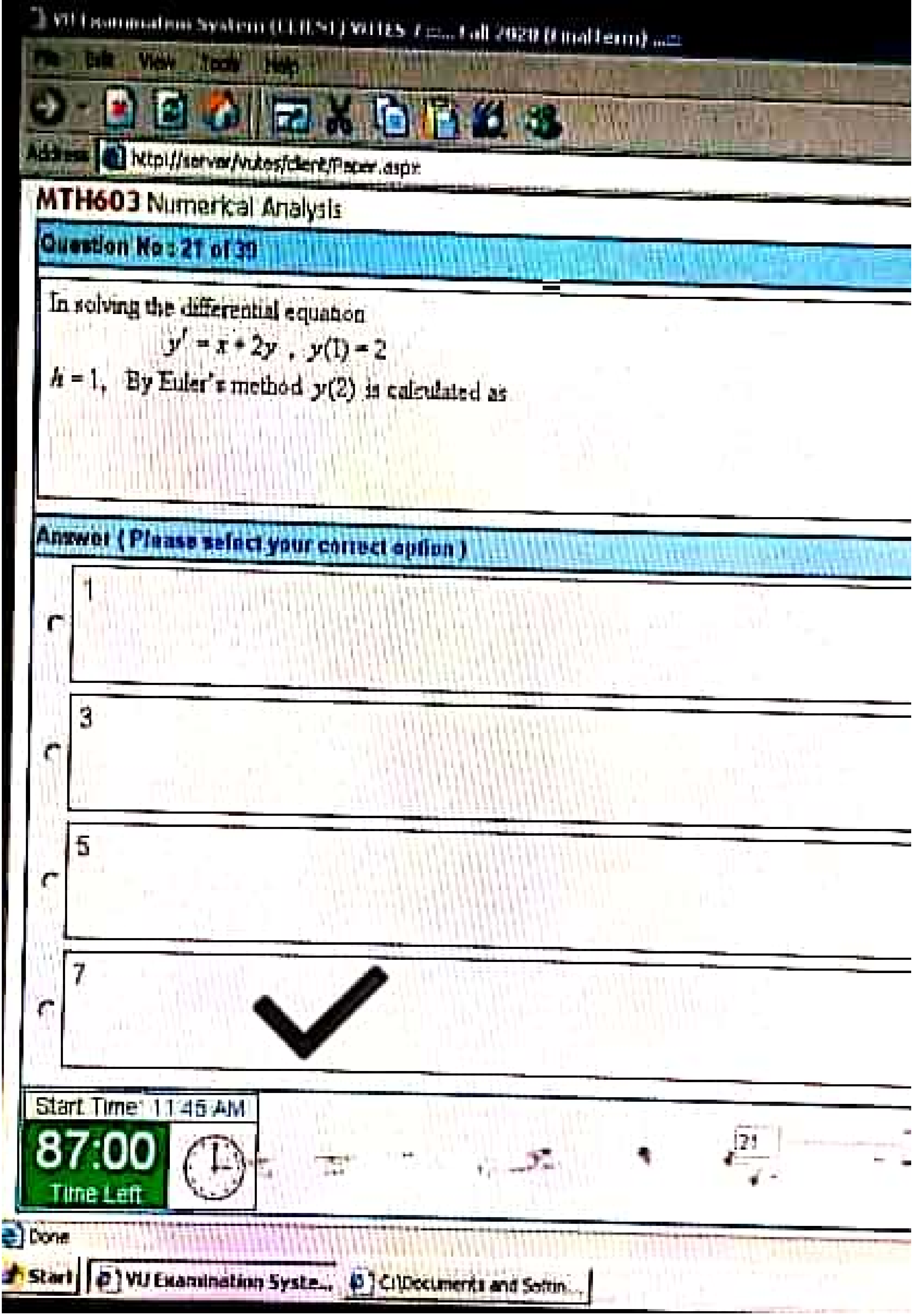
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MTH603 Numerical Analysis

Question No : 21 of 30

In solving the differential equation

$$y' = x + 2y \ , \ y(1) = 2$$

$h = 1$, By Euler's method $y(2)$ is calculated as

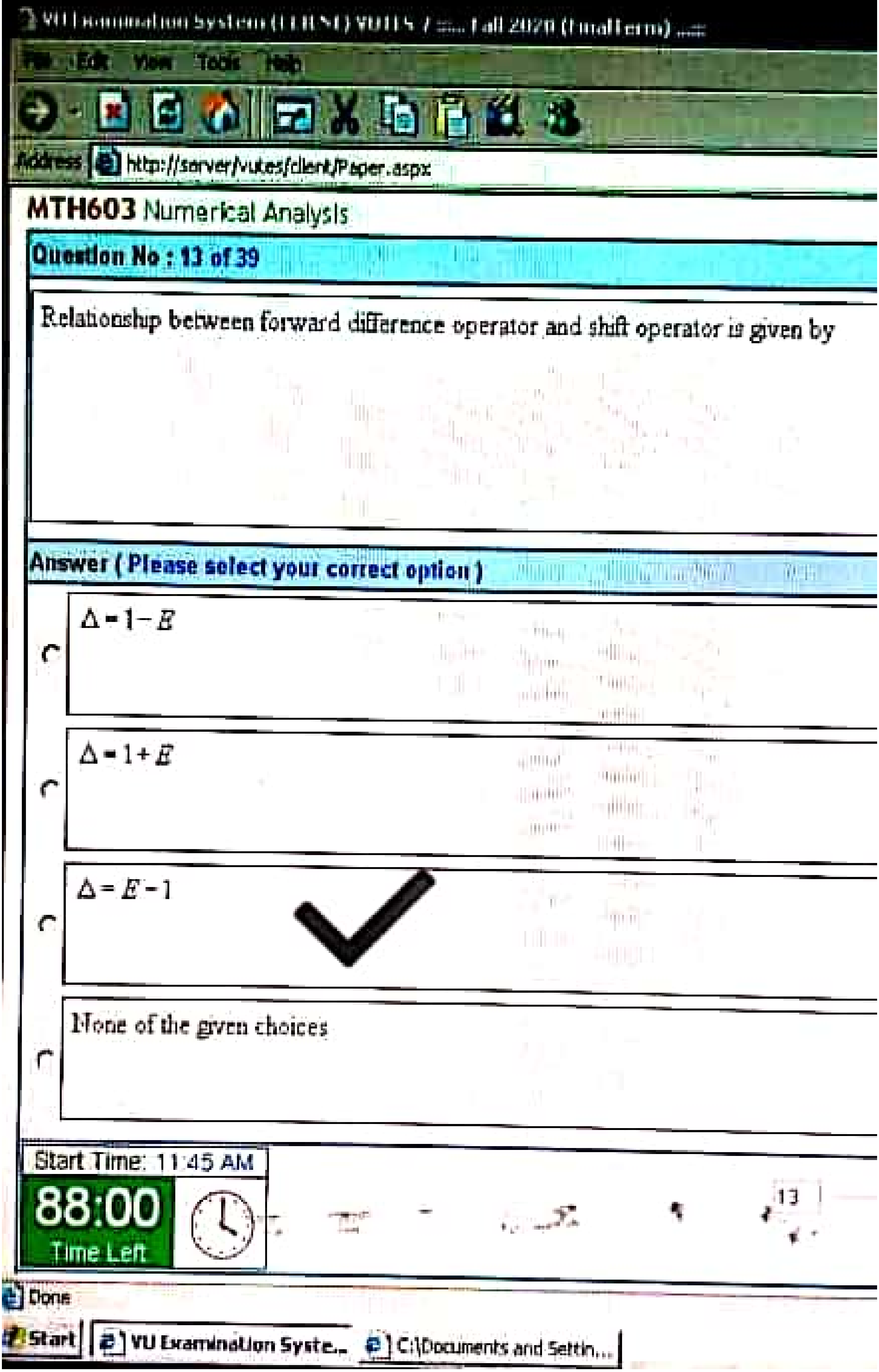
Answer (Please select your correct option)

○ 1

○ 3

○ 5

[✓ ○ 7]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MTH603 Numerical Analysis

Question No : 13 of 39

Relationship between forward difference operator and shift operator is given by

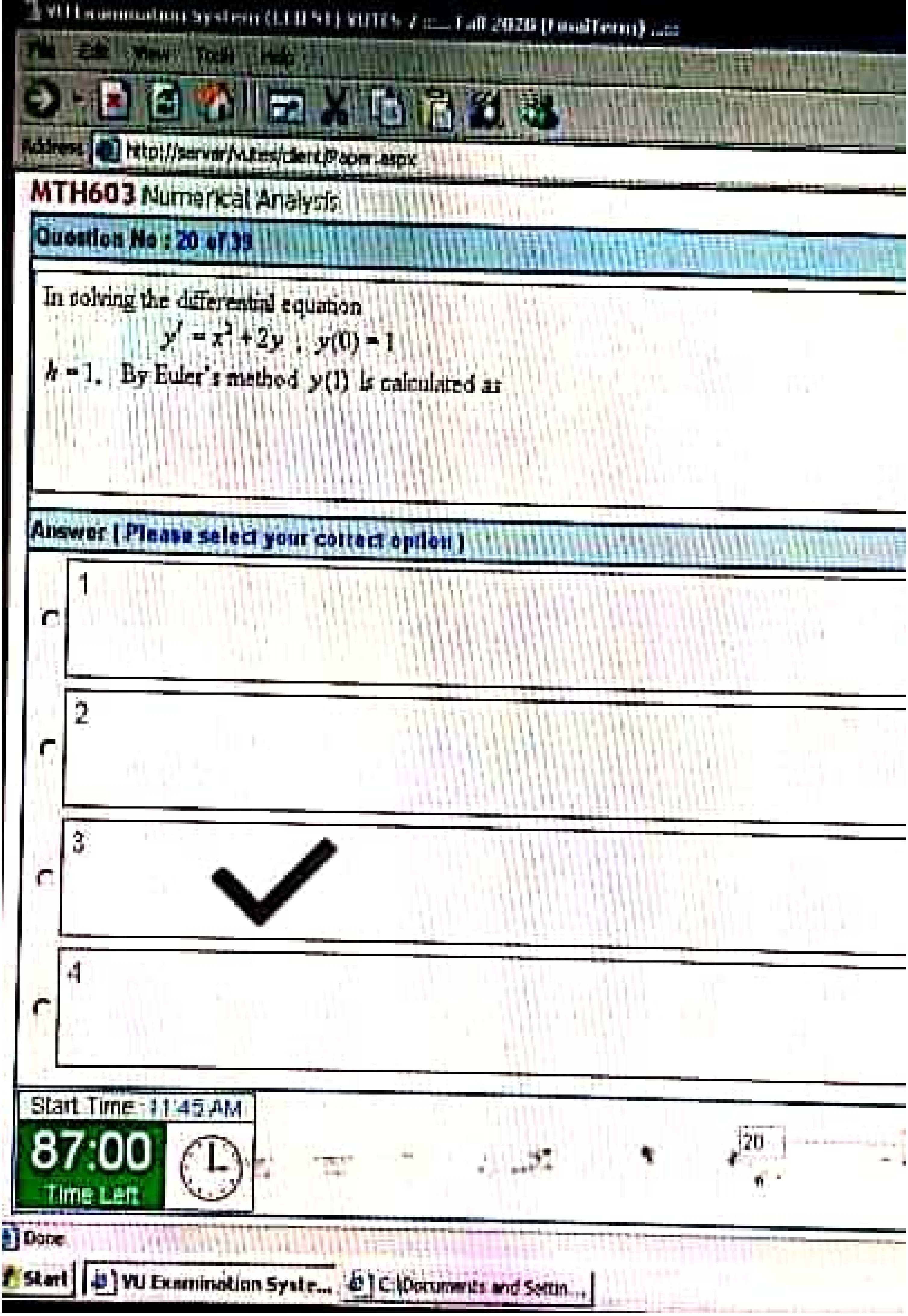
Answer (Please select your correct option)

○ $\Delta = 1 - E$

○ $\Delta = 1 + E$

○ [✓ $\Delta = E - 1$]

○ None of the given choices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MTH603 Numerical Analysis

Question No : 20 of 39

In solving the differential equation

$$y' = x^2 + 2y \ ; \ y(0) = 1$$

$h = 1$. By Euler's method $y(1)$ is calculated as

Answer (Please select your correct option)

○ 1

○ 2

○ [✓ 3]

○ 4

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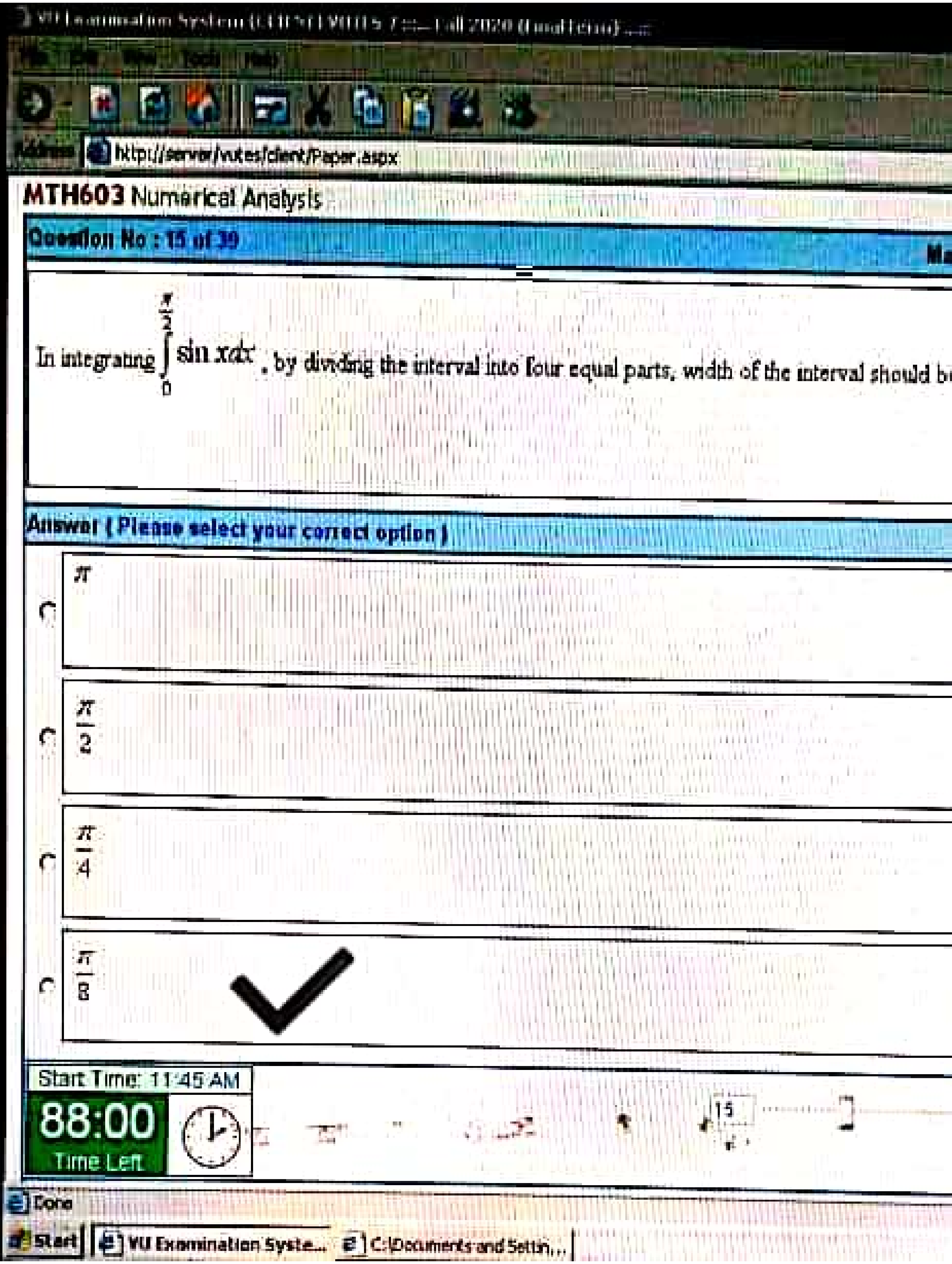

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MTH603 Numarical Analysis

Question No : 15 of 39

In integrating $\int_0^{\frac{\pi}{2}} \sin x dx$, by dividing the interval into four equal parts, width of the interval should b

Answer (Please select your correct option)

○ π

○ $\frac{\pi}{2}$

○ $\frac{\pi}{4}$

○ $\frac{\pi}{8}$ ✓

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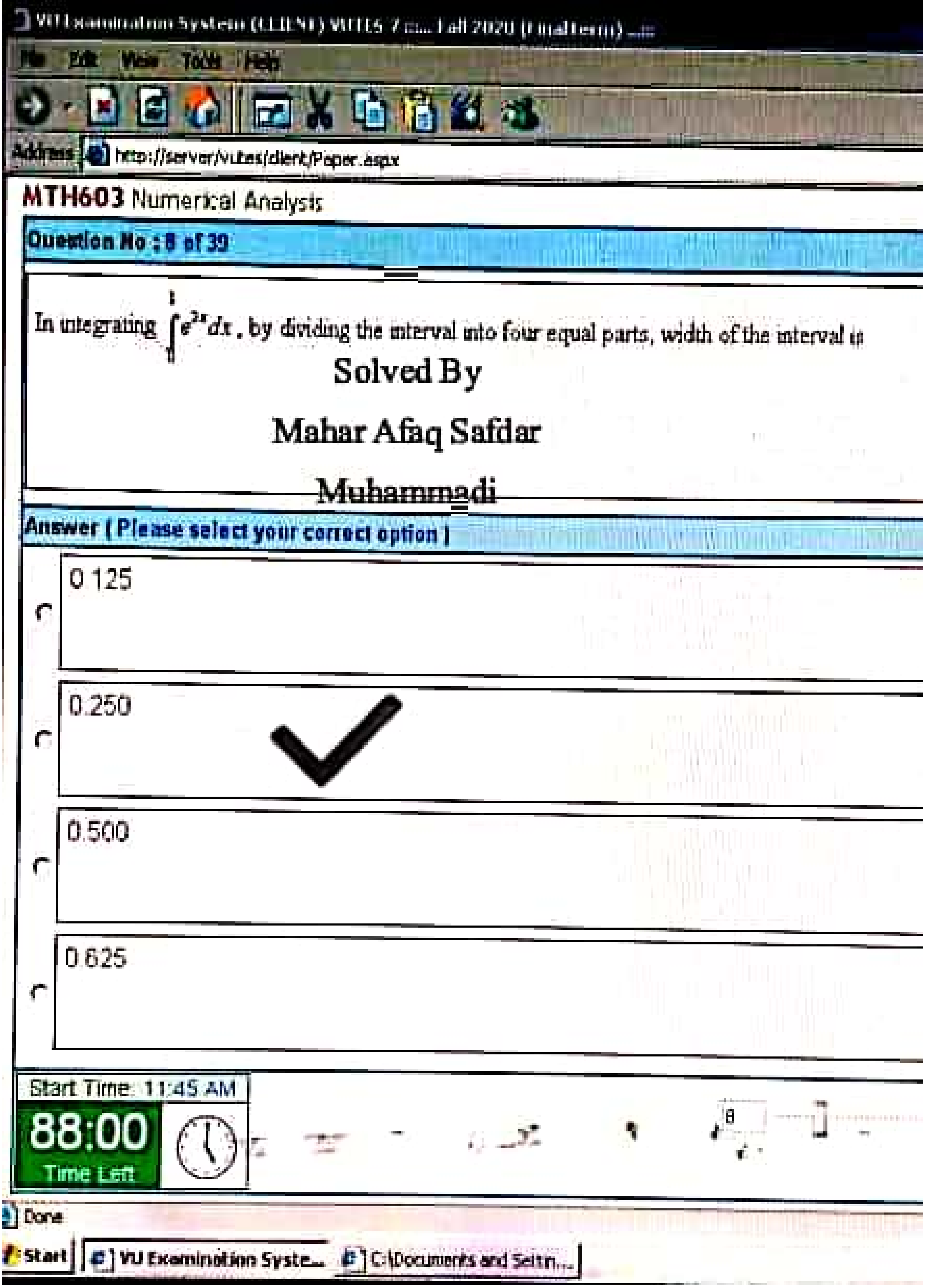
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MTH603 Numerical Analysis

Question No : 8 of 39

In integrating $\int_0^1 e^{2x}\,dx$, by dividing the interval into four equal parts, width of the interval is

Solved By

Mahar Afaq Safdar

Muhammadi

Answer (Please select your correct option)

○ 0.125

○ [✓ 0.250]

○ 0.500

○ 0.625

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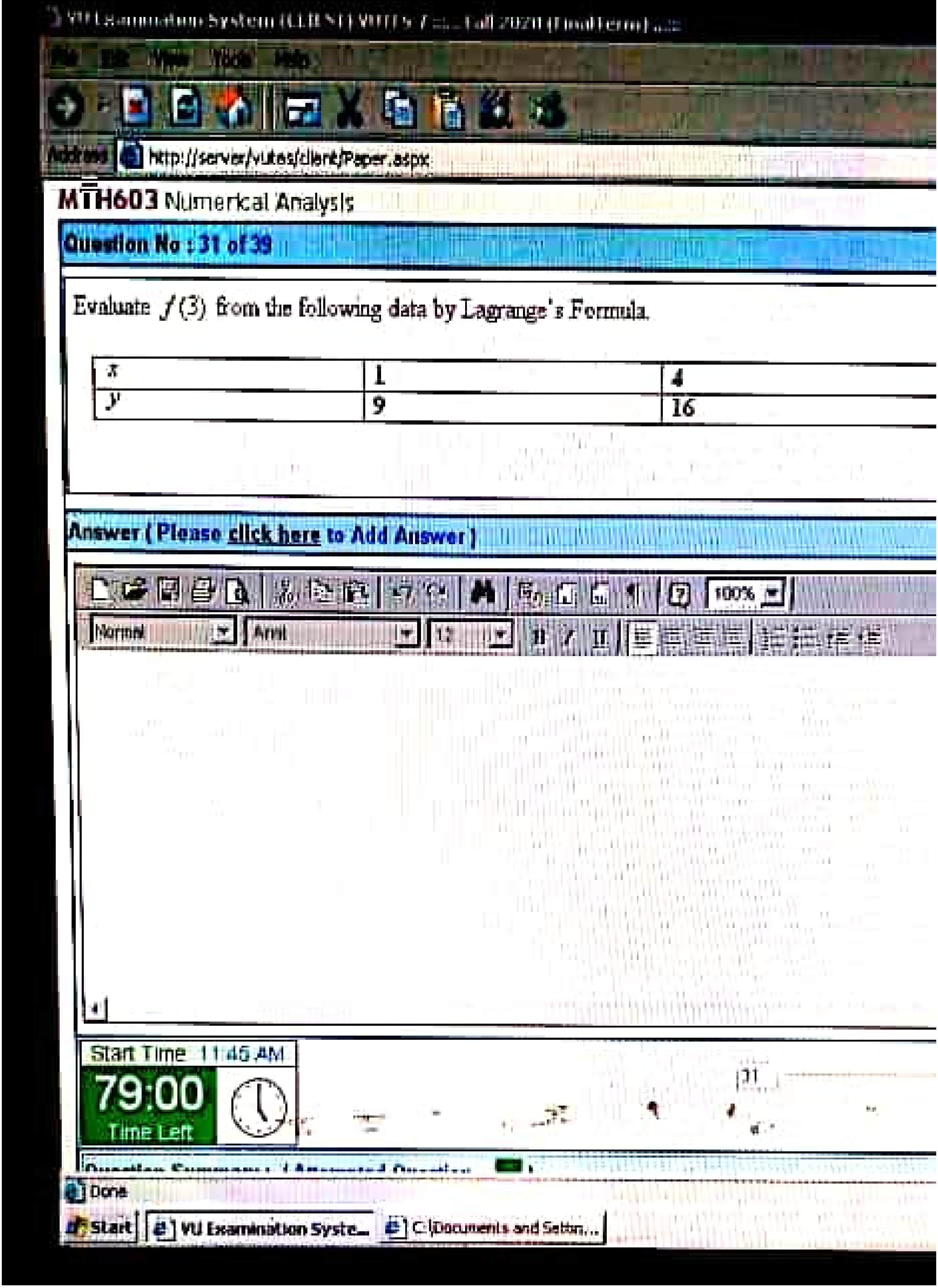
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MTH603 Numerical Analysis

Question No : 31 of 39

Evaluate $f(3)$ from the following data by Lagrange's Formula.

x	1	4
y	9	16

Answer (Please click here to Add Answer)

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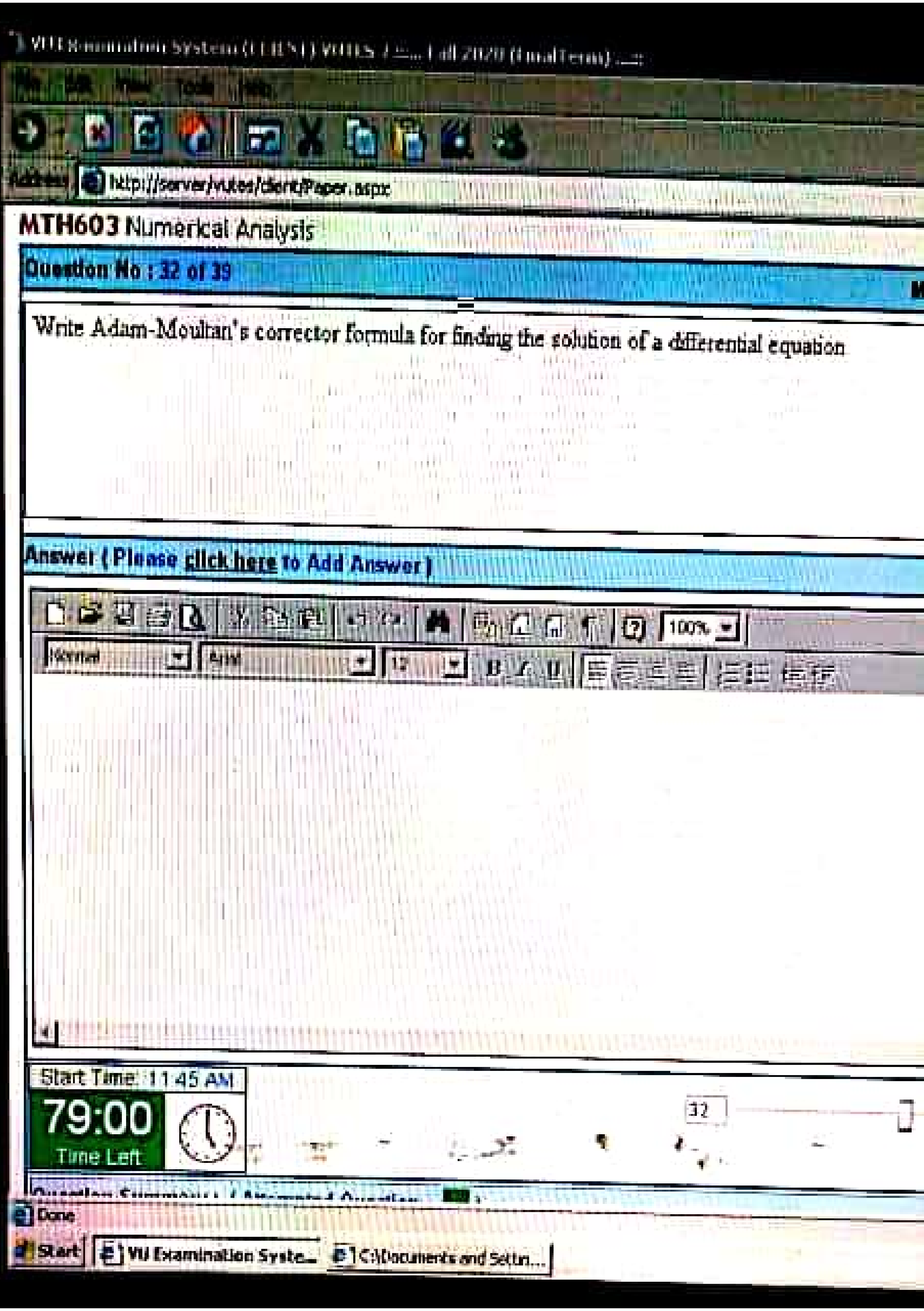
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MTH603 Numerical Analysis

Question No : 32 of 39

Write Adam-Moultan's corrector formula for finding the solution of a differential equation

Answer (Please click here to Add Answer)

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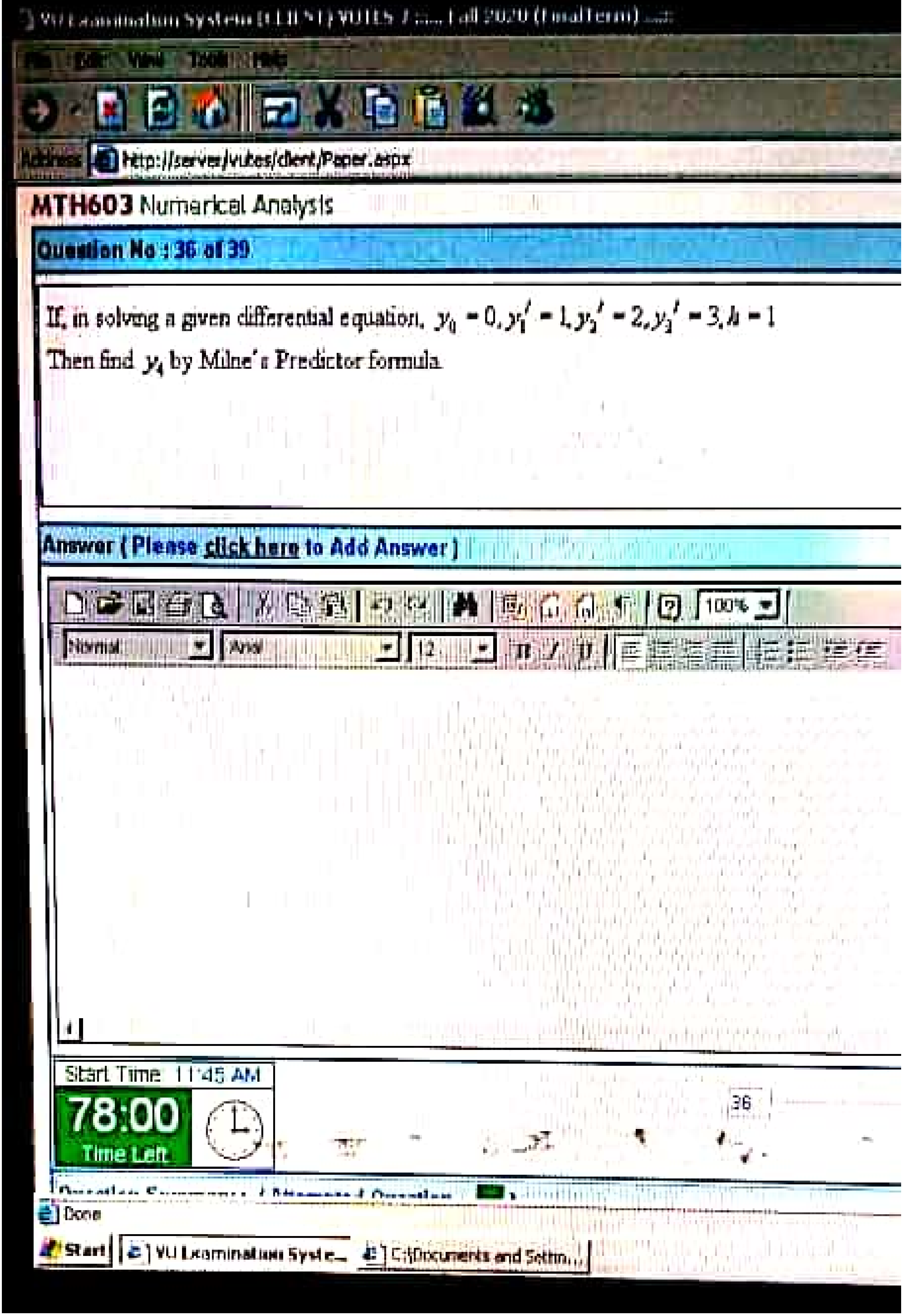
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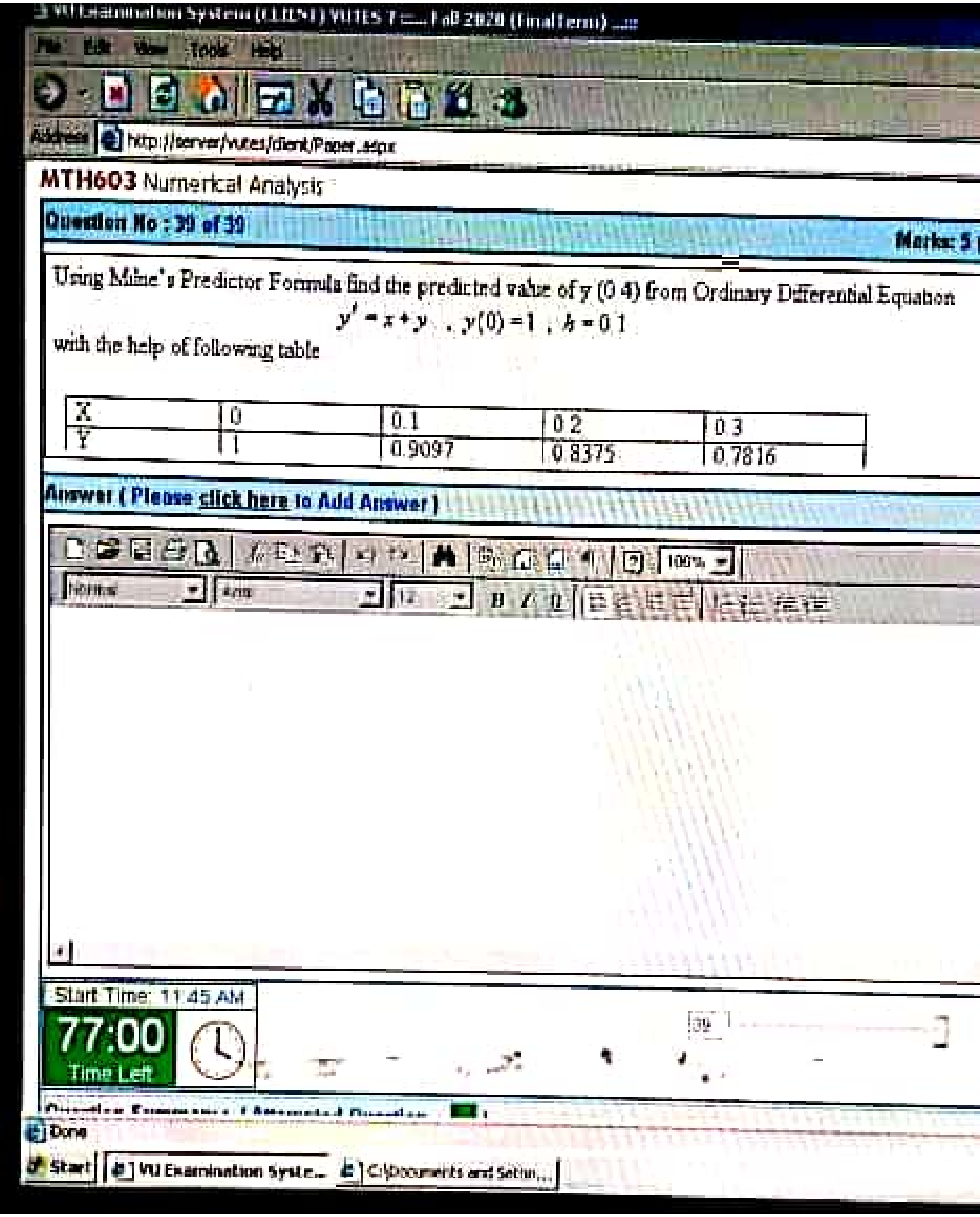
MTH603 Numerical Analysis

Question No : 36 of 39

If, in solving a given differential equation, $y_0 = 0, y_1' = 1, y_2' = 2, y_3' = 3, h = 1$

Then find y_4 by Milne's Predictor formula.

Answer (Please click here to Add Answer)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MTH603 Numerical Analysis

Question No : 39 of 39

Marks: 5

Using Milne's Predictor Formula find the predicted value of y (0.4) from Ordinary Differential Equation

$$y' = x + y \quad , y(0) = 1 \; , h = 0.1$$

with the help of following table

X	0	0.1	0.2	0.3
Y	1	0.9097	0.8375	0.7816

Answer (Please click here to Add Answer)

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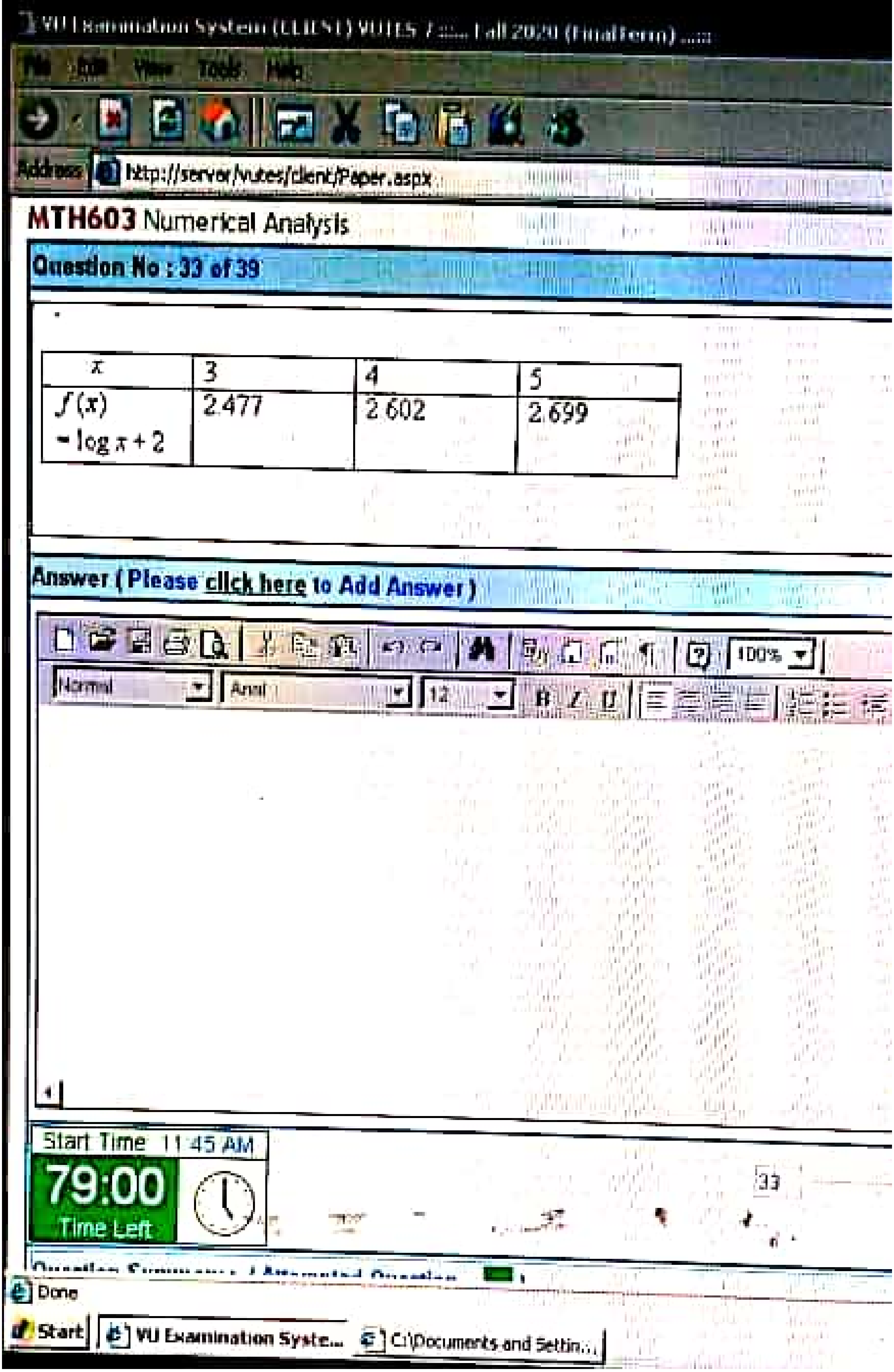
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MTH603 Numerical Analysis

Question No : 33 of 39

x	3	4	5
$f(x) = \log x + 2$	2.477	2.602	2.699

Answer (Please click here to Add Answer)

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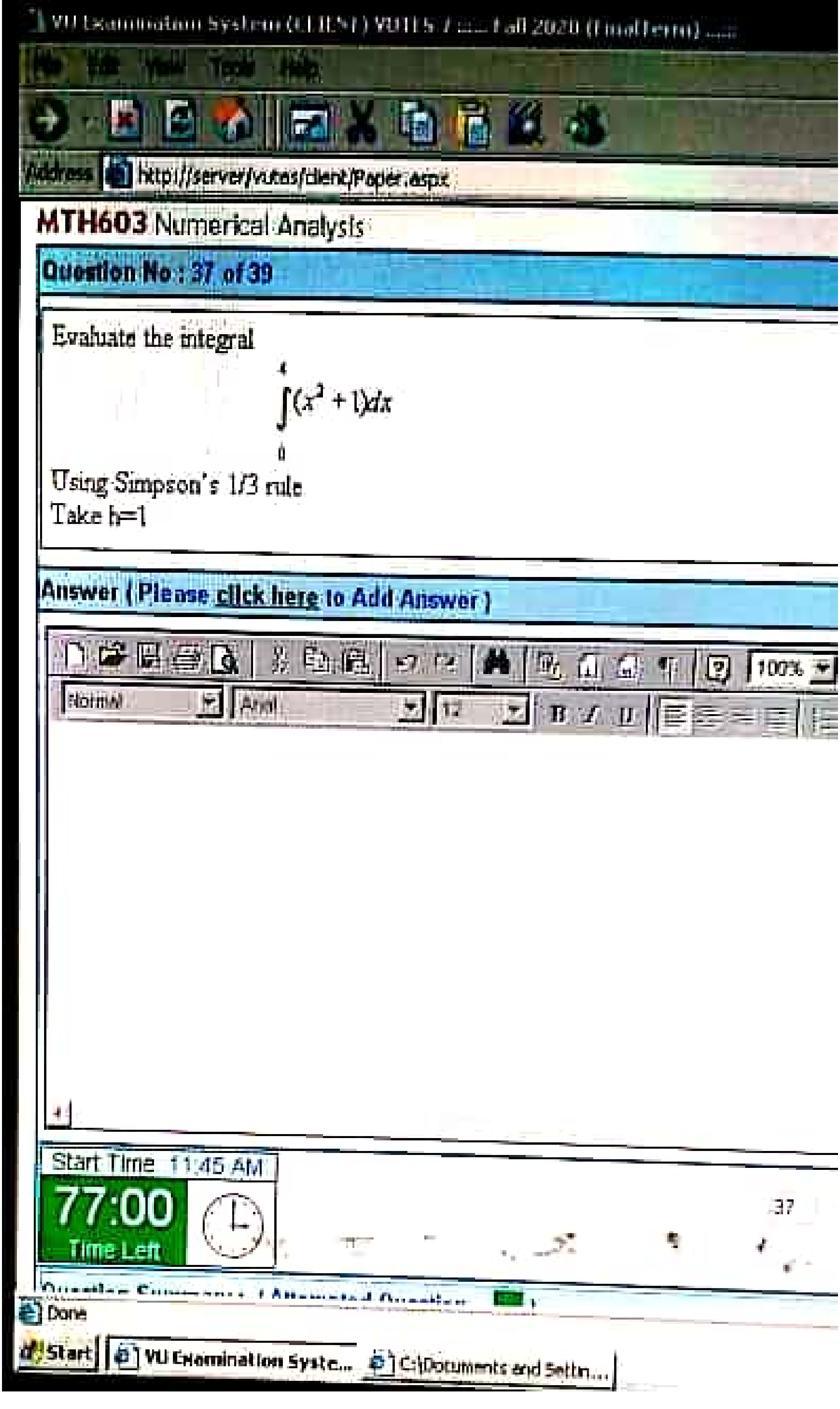
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MTH603 Numerical Analysis

Question No : 37 of 39

Evaluate the integral

$$\int_0^4 (x^2+1)dx$$

Using Simpson's 1/3 rule
Take h=1

Answer (Please click here to Add Answer)

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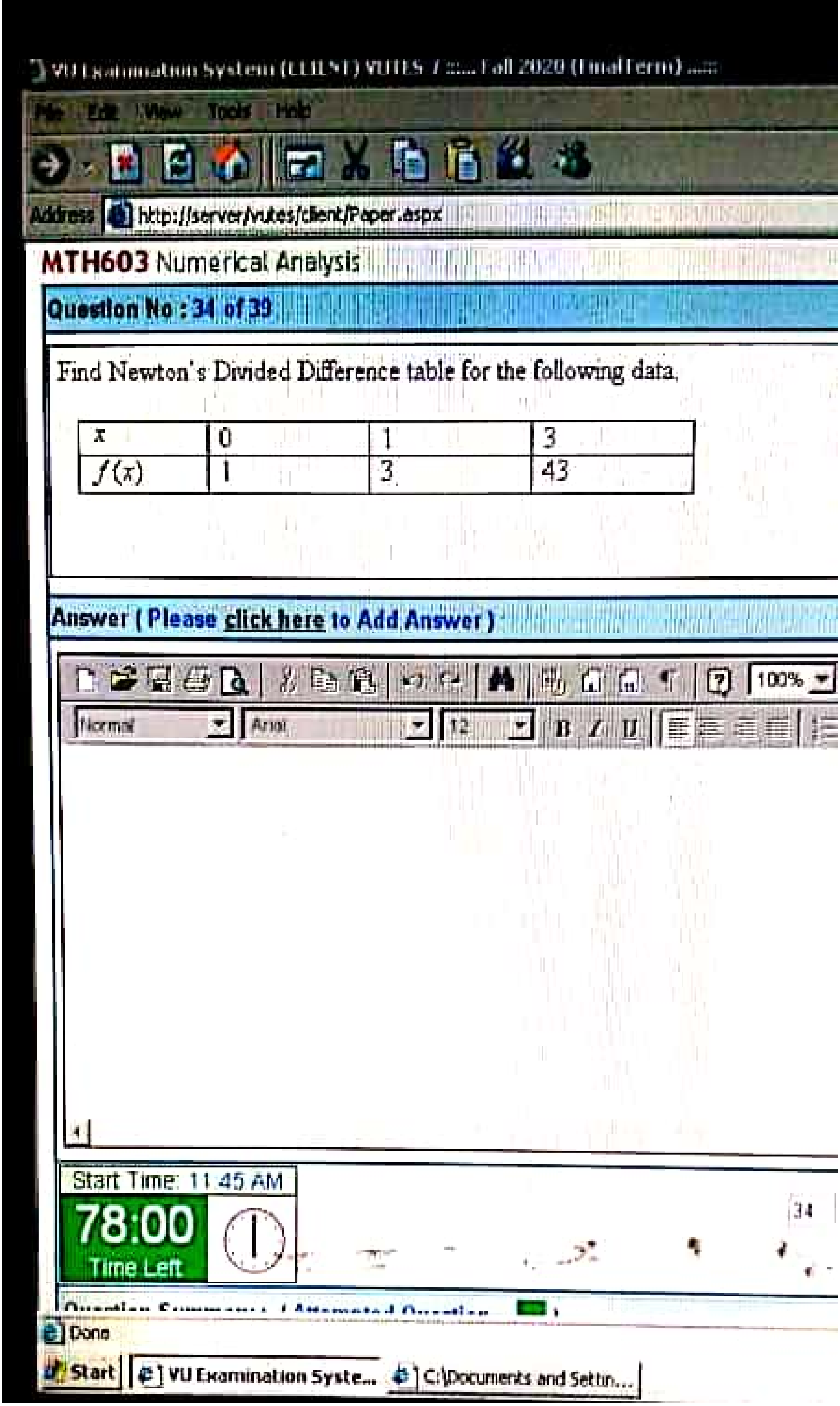
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MTH603 Numerical Analysis

Question No : 34 of 39

Find Newton's Divided Difference table for the following data.

x	0	1	3
$f(x)$	1	3	43

Answer (Please click here to Add Answer)

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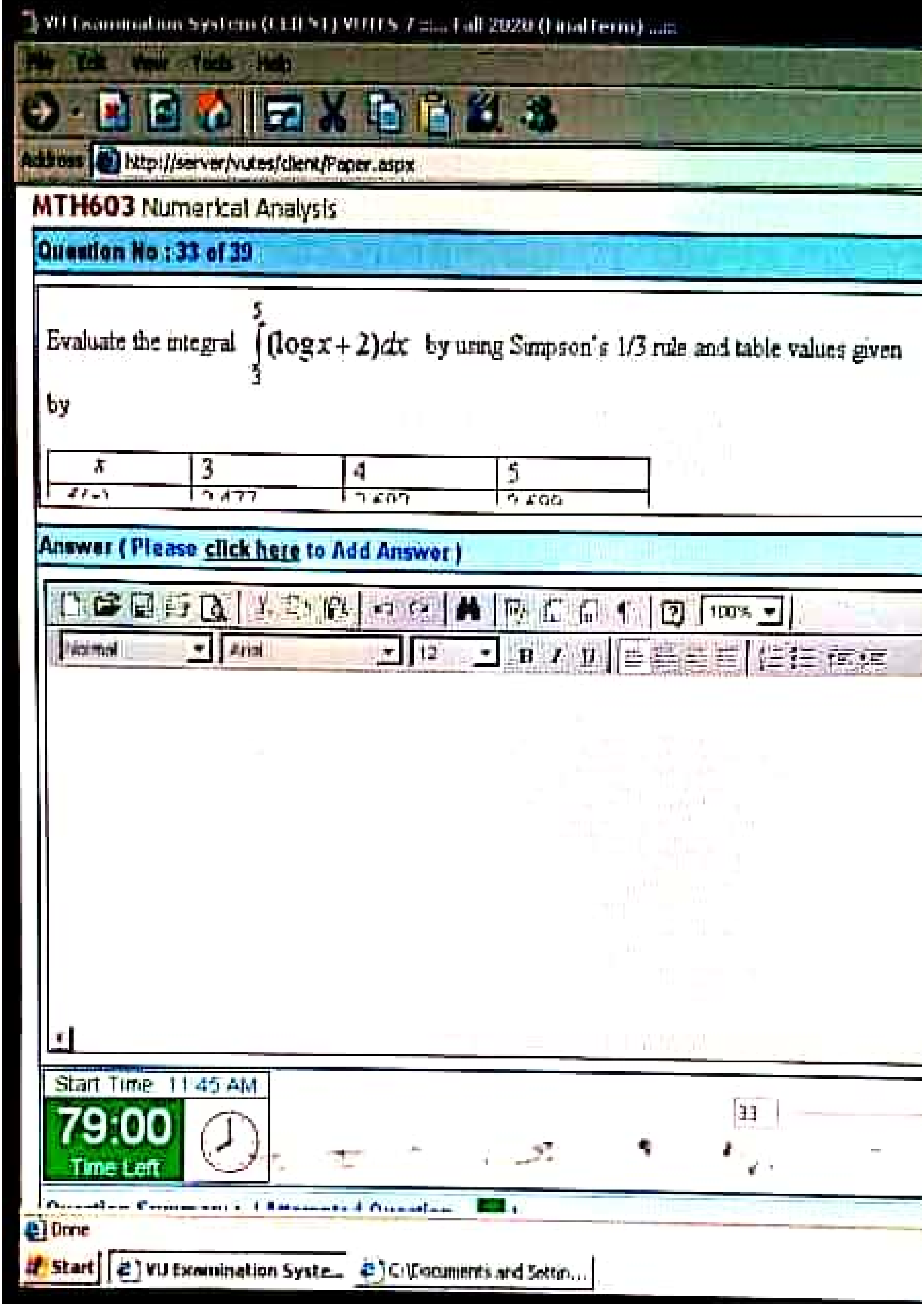
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MTH603 Numerical Analysis

Question No : 33 of 39

Evaluate the integral $\int_{3}^{5}(\log x+2)dx$ by using Simpson's 1/3 rule and table values given by

x	3	4	5
$f(x)$	2.477	2.602	2.699

Answer (Please click here to Add Answer)

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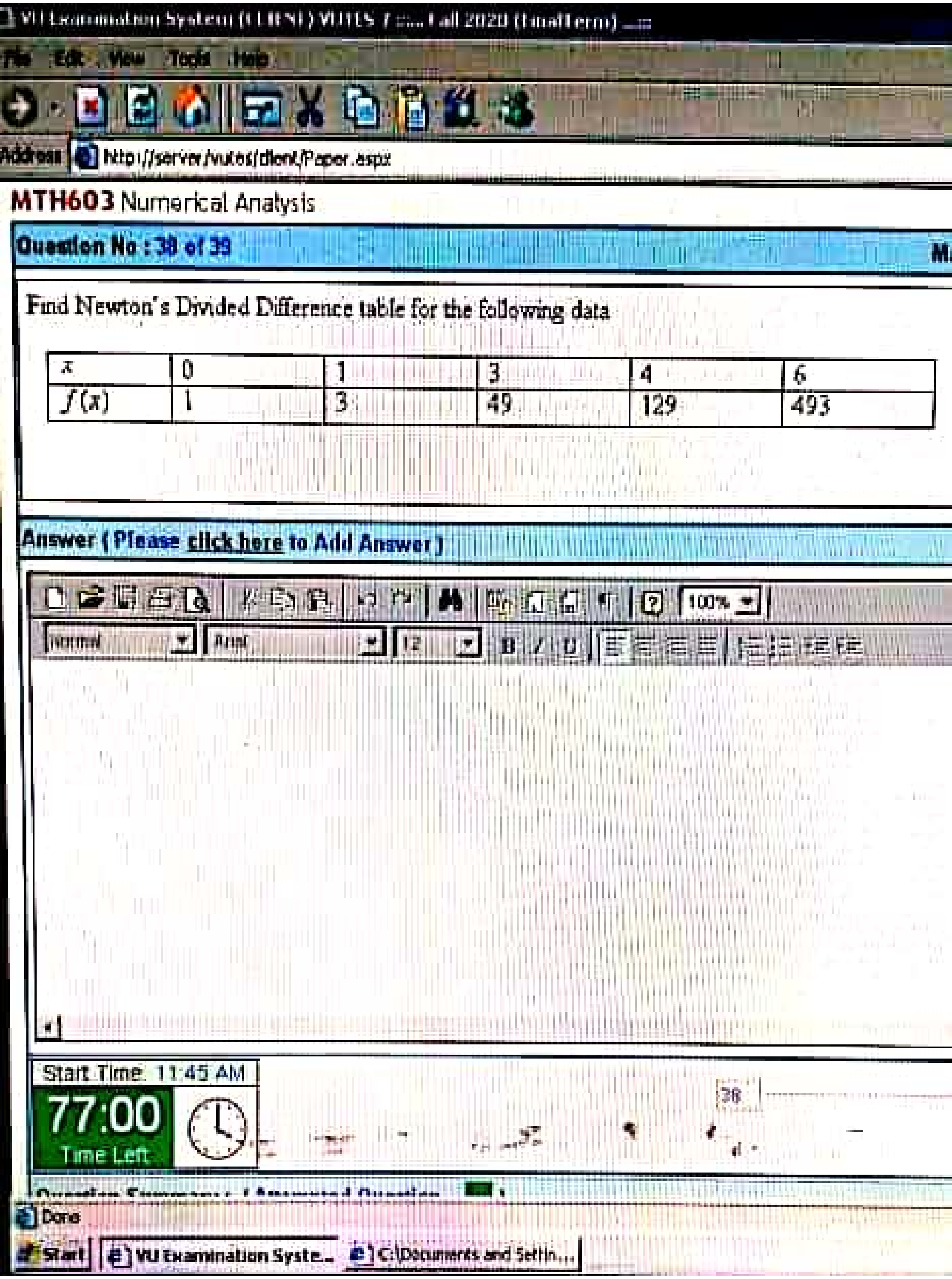
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MTH603 Numerical Analysis

Question No : 38 of 39

Find Newton's Divided Difference table for the following data

x	0	1	3	4	6
$f(x)$	1	3	49	129	493

Answer (Please click here to Add Answer)

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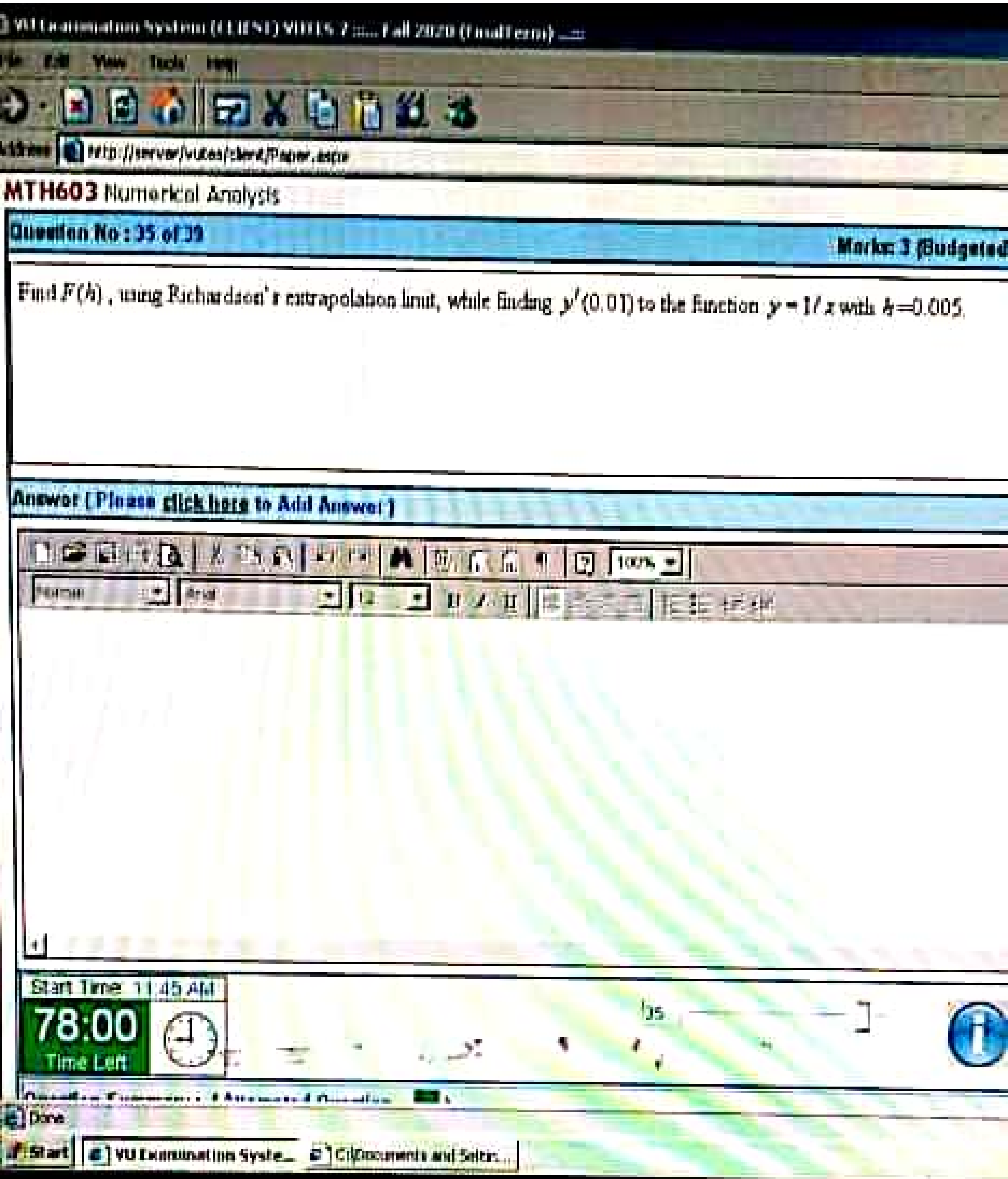

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MTH603 Numerical Analysis

Question No : 35 of 35 **Marks: 3 (Budgeted**

Find $F(h)$, using Richardson's extrapolation limit, while finding $y'(0.01)$ to the function $y = 1/x$ with $h=0.005$.